



TEST REPORT

Applicant: ZHEJIANG QIDI TECHNOLOGY CO.,LTD

Address: No.709 Huaguang Road, Tingtian Street, Wenzhou City, Zhejiang Province

Manufacturer: ZHEJIANG QIDI TECHNOLOGY CO.,LTD

Address: No.709 Huaguang Road, Tingtian Street, Wenzhou City, Zhejiang Province

Product Name: 3D printer

Trade Mark: N/A

Model Number: Max4, Plus5

Date of Receipt: Nov. 19, 2025

Test Date: Nov. 19, 2025 - Dec. 04, 2025

Date of Report: Dec. 04, 2025

Prepared By: Shenzhen DL Testing Technology Co., Ltd.

Address: 101-201, Comprehensive Building, Tongzhou Electronics Longgang Factory Area, No.1 Baolong Fifth Road, Baolong Community, Baolong Street, Longgang District, Shenzhen, China

Applicable Standards: ETSI EN 301 489-1 V2.2.3 (2019-11),
ETSI EN 301 489-17 V3.3.1 (2024-09)
EN 55032:2015+A1:2020, EN 55035:2017+A11:2020

Test Result: Pass

Report Number: DLE-251119036-2R

Prepared (Engineer): Ken Tan

Reviewer (Supervisor): Jack Bu

Approved (Manager): Jade Yang



This test report is based on a single evaluation of one sample of above mentioned products. It is not permitted to be duplicated in extracts without written approval of Shenzhen DL Testing Technology Co., Ltd.

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1. VERSION

Version No.	Date	Description
00	Dec. 04, 2025	Original

2. TEST SUMMARY

EMC Emission				
Standard	Test Item	Limit	Result	Remark
ETSI EN 301 489-1, EN 55032	Conducted Emission at power ports	Class B	PASS	
	Conducted Emission at LAN port	Class B	N/A	
	Radiated Emission below 1GHz	Class B	PASS	
	Radiated Emission above 1GHz	Class B	PASS	
EN IEC 61000-3-2	Harmonic Current Emission	Class A or D	PASS	
EN 61000-3-3	Voltage Fluctuations & Flicker	-----	PASS	
EMC Immunity				
Section ETSI EN 301 489-17, EN 55035	Test Item	Performance Criteria	Result	Remark
EN 61000-4-2	Electrostatic Discharge	B	PASS	
EN IEC 61000-4-3	RF electromagnetic field	A	PASS	
EN 61000-4-4	Fast transients	B	PASS	
EN 61000-4-5	Surges	B	PASS	
EN IEC 61000-4-6	Injected Current	A	PASS	
EN IEC 61000-4-11	Volt. Interruptions Volt. Dips	B / C / C	PASS	

NOTE:

(1)" N/A" denotes test is not applicable in this Test Report

(4) Test Facility: Shenzhen DL Testing Technology Co., Ltd.

Address: 101-201, Comprehensive Building, Tongzhou Electronics Longgang Factory Area, No.1
Baolong Fifth Road, Baolong Community, Baolong Street, Longgang District, Shenzhen, China



3. GENERAL INFORMATION

3.1 Description of Device (EUT)

EUT: 3D printer

Trade Mark: N/A

Model Number: Max4, Plus5

Test Model: Max4

Model difference: All models are same as the samples except model name and appearance color, they have the same structure and circuit.

Power Supply: 100-240V ~ 50/60Hz 3.4A Max

Work Frequency: Above 108MHz

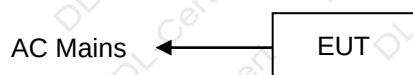
Note: (1) For a more detailed features description, please refer to the manufacturer's specifications or the User's Manual.

(2) The EUT's all information provided by client.

3.2 Tested System Details

None.

3.3 Block Diagram of Test Set-up



3.4 Test Mode Description

Mode1.: On Mode

3.5 Test Auxiliary Equipment

None.

3.6 Test Uncertainty

Conducted Emission Uncertainty : $\pm 2.56\text{dB}$

Radiated Emission Uncertainty : $\pm 3.24\text{dB}$

**4. TEST INSTRUMENT USED****For Conducted Emission Test (843 Shielded Room)**

Equipment	Manufacturer	Model	Serial	Last Cal.	Next Cal.
843 Shielded Room	YIHENG	843 Room	843	Nov. 05, 2023	Nov. 04, 2026
EMI Receiver	R&S	ESR	101421	Oct. 25, 2025	Oct. 24, 2026
LISN	R&S	ENV216	102417	Oct. 25, 2025	Oct. 24, 2026
Clamp	COM-POWER	CLA-050	431072	Oct. 27, 2025	Oct. 26, 2026
3-Loop Antenna	DAZE	ZN30401	13021	Oct. 27, 2025	Oct. 26, 2026
ISN T8	Schwarzbeck	NTFM 8158	101135	Oct. 25, 2025	Oct. 24, 2026
ISN T5	Schwarzbeck	NTFM 8158	101136	Oct. 25, 2025	Oct. 24, 2026
843 Cable 1#	ChengYu	CE Cable	001	Oct. 25, 2025	Oct. 24, 2026
843 Cable 1#	ChengYu	CL Cable	002	Oct. 25, 2025	Oct. 24, 2026

For Radiated Emission Test (966 chamber)

Equipment	Manufacturer	Model	Serial	Last Cal.	Next Cal.
966 chamber	YIHENG	966 Room	966	Nov. 06, 2023	Nov. 05, 2026
Spectrum Analyzer	Agilent	E4408B	MY50140780	Oct. 25, 2025	Oct. 24, 2026
EMI Receiver	R&S	ESRP7	101393	Oct. 25, 2025	Oct. 24, 2026
Amplifier	Schwarzbeck	BBV9743B	00153	Oct. 25, 2025	Oct. 24, 2026
Amplifier	EMEC	EM01G8GA	00270	Oct. 25, 2025	Oct. 24, 2026
Broadband Trilog Antenna	Schwarzbeck	VULB9162	00306	Oct. 27, 2025	Oct. 26, 2026
Horn Antenna	Schwarzbeck	BBHA9120D	02139	Oct. 27, 2025	Oct. 26, 2026
966 Cable 1#	ChengYu	966	004	Oct. 25, 2025	Oct. 24, 2026
966 Cable 2#	ChengYu	966	003	Oct. 25, 2025	Oct. 24, 2026

For Harmonic & Flicker Test (EMS --- site)

Equipment	Manufacturer	Model	Serial	Last Cal.	Next Cal.
Harmonics, Flicker & power Analyser	LAPLACE INSTRUMENTS	AC2000A	311370	Oct. 25, 2025	Oct. 24, 2026
AC Power Supply	MToni	HPF5010	633659	Oct. 25, 2025	Oct. 24, 2026

For Electrostatic Discharge Immunity Test (EMS --- site)

Equipment	Manufacturer	Model	Serial	Last Cal.	Next Cal.
ESD Tester	SCHLODER	SESD 230	17352	Oct. 27, 2025	Oct. 26, 2026

**For RF Field Strength Susceptibility Test (Other RS site)**

Equipment	Manufacturer	Model	Serial	Last Cal.	Next Cal.
Signal Generator	HP	SMB100A	1406600K02	Oct. 14, 2025	Oct. 13, 2026
Amplifier	BONN ELEKTRONIK	BLWAO810-250/100D	066454	Oct. 14, 2025	Oct. 13, 2026
Amplifier	BONN ELEKTRONIK	BLWA0840-5030D	066453	Oct. 14, 2025	Oct. 13, 2026
RS Test Antenna	SKET	STLP 9129 Plus	202301/428	Oct. 14, 2025	Oct. 13, 2026
audio analyzer	R&S	UPL	16755	Oct. 14, 2025	Oct. 13, 2026
Antenna	EMCO	3108	9507-2534	Oct. 14, 2025	Oct. 13, 2026
Log-periodic Antenna	A&R	AT1080	16812	Oct. 14, 2025	Oct. 13, 2026
Power Sensor	R&S	Z11	116655	Oct. 14, 2025	Oct. 13, 2026
Power Sensor	R&S	Z11	121896	Oct. 14, 2025	Oct. 13, 2026
Horn Antenna	AINFO	JXTXLB-10180-SF	J2031090903006	Oct. 14, 2025	Oct. 13, 2026

For EFT /B, Surge, Voltage Dips Interruptions Test (EMS --- site)

Equipment	Manufacturer	Model	Serial	Last Cal.	Next Cal.
Transient Comprehensive Immunity Test System	Graphtec	HVIP16T+HCO MPACT 5	192501+192202	Oct. 25, 2025	Oct. 24, 2026
Coupling Clamp	HTEC	001	0001	Oct. 25, 2025	Oct. 24, 2026

For Injected Currents Susceptibility Test (EMS --- site)

Equipment	Manufacturer	Model	Serial	Last Cal.	Next Cal.
C/S Test System	LIONCEL	RIS-6091-85	0191101	Oct. 25, 2025	Oct. 24, 2026
CDN	LIONCEL	CDN-M2-16	0191001	Oct. 25, 2025	Oct. 24, 2026
CDN	LIONCEL	CDN-M3-16	0191002	Oct. 25, 2025	Oct. 24, 2026
Injection Clamp	Frankonia	EMCL-20	18101728-0108	Oct. 25, 2025	Oct. 24, 2026
Attenuator	LIONCEL	100W 6dB DC-3GHz	0191003	Oct. 25, 2025	Oct. 24, 2026

Other

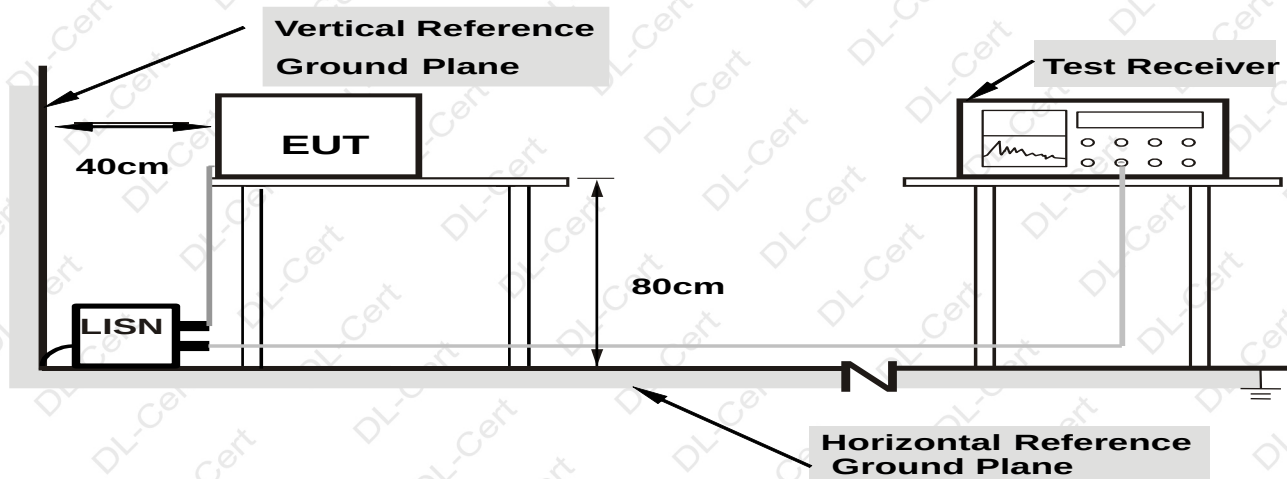
Item	Name	Manufacturer	Model	Software version
1	EMC Conduction Test System	FALA	EZ EMC	EMC-CON 3A1.1
2	EMC radiation test system	FALA	EZ EMC	FA-03A2
3	Harmonic test system	LAPLACE INSTRUMENTS	Harmonic	1.0.0.0
4	RF Immunity test system	LIONCEL	RF Immunity	1.8.0



5. CONDUCTED EMISSION TEST

5.1 Block Diagram of Test Setup

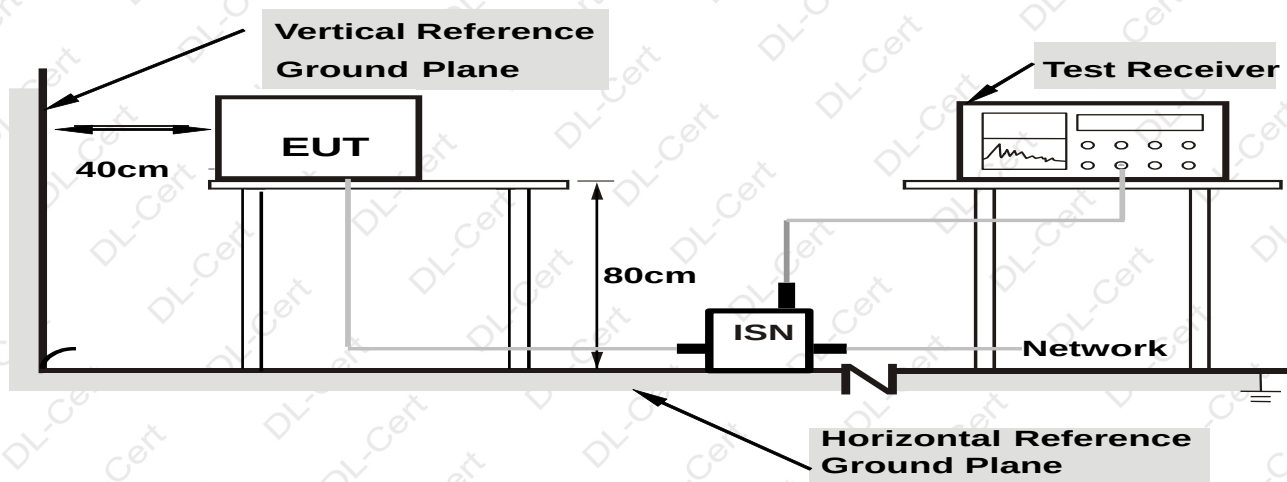
For Mains Terminals Test



Note: 1.Support units were connected to second LISN.

2.Both of LISNs (AMN) are 80 cm from EUT and at least 80 from other units and other metal planes

For Telecom Port Test



Note: 1.Support units were connected to second LISN.

2.Both of ISNs are 80 cm from EUT and at least 80 cm from other units and other metal planes



5.2 Test Standard and Limit

EN 55032

For Mains Terminals Test			For Telecom Port Test		
Frequency MHz	Limits dB(μ V)		Frequency MHz	Limits dB(μ V)	
	Quasi-peak Level	Average Level		Quasi-peak Level	Average Level
0.15~0.50	66 ~ 56*	56 ~ 46*	0.15~0.50	84 ~ 74*	74 ~ 64*
0.50~5.00	56	46	0.50~30.00	74	64
5.00~30.00	60	50	/	/	/

Notes: 1. *Decreasing linearly with logarithm of frequency.

2. The lower limit shall apply at the transition frequencies.

5.3 EUT Configuration on Test

The following equipment's are installed on conducted emission test to meet EN 55032 requirement and operating in a manner which tends to maximize its emission characteristics in a normal application.

5.4 Operating Condition of EUT

5.4.1 Setup the EUT and simulators as shown in Section 5.1.

5.4.2 Turn on the power of all equipment.

5.4.3 Let the EUT work in test modes and test it.

5.5 Test Procedure

The EUT is put on the table and connected to the AC mains through a Artificial Mains Network (AMN) or ISN. This provided a 50ohm coupling impedance for the tested equipment's. Both sides of AC line are checked to find out the maximum conducted emission levels according to the **EN 55032** regulations during conducted emission test.

The bandwidth of the test receiver (R&S Test Receiver ESR) is set at 10KHz.

The frequency range from 150 KHz to 30 MHz is investigated.

5.6 Test Result

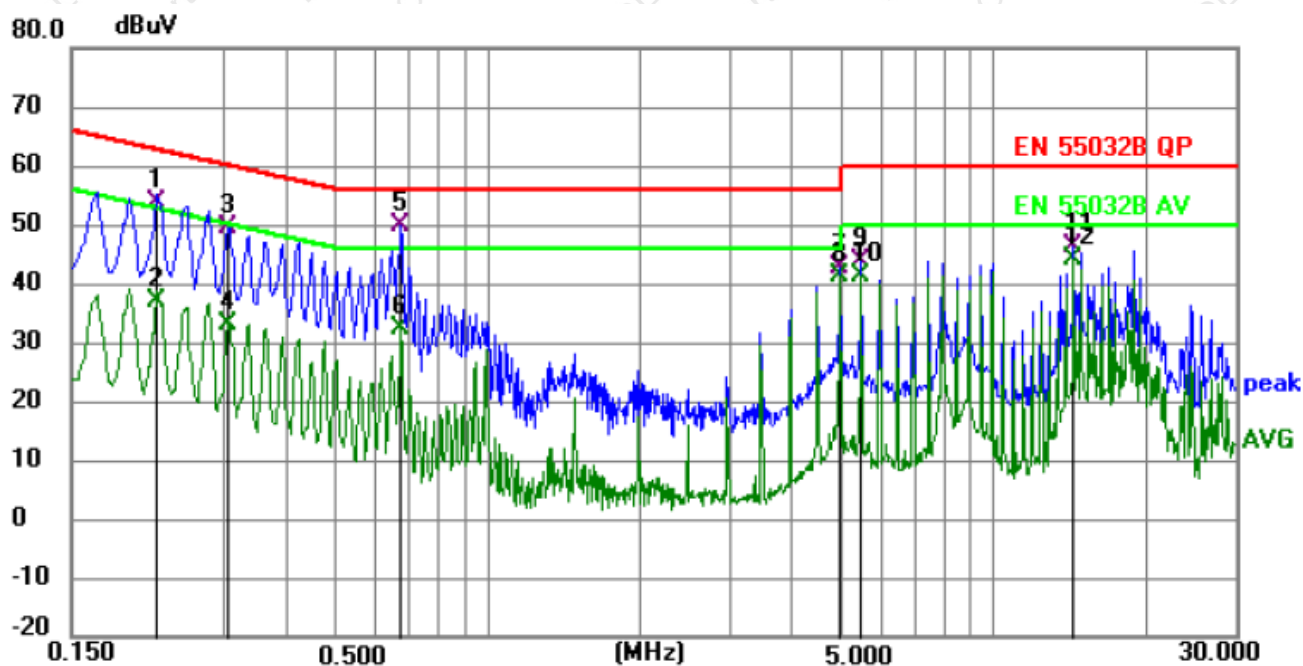
PASS

Please refer to the following page.



Conducted Emission Test Data

Temperature:	24.5 °C	Relative Humidity:	54%
Pressure:	1009hPa	Phase:	Line
Test Voltage:	AC 230V/50Hz	Test Mode:	Mode 1



No.	Frequency (MHz)	Reading (dBuV)	Factor (dB)	Level (dBuV)	Limit (dBuV)	Margin (dB)	Detector	P/F	Remark
1	0.2220	43.74	9.94	53.68	62.74	-9.06	QP	P	
2	0.2220	26.86	9.94	36.80	52.74	-15.94	AVG	P	
3	0.3075	39.44	9.92	49.36	60.04	-10.68	QP	P	
4	0.3075	23.04	9.92	32.96	50.04	-17.08	AVG	P	
5	0.6720	39.78	9.95	49.73	56.00	-6.27	QP	P	
6	0.6720	22.30	9.95	32.25	46.00	-13.75	AVG	P	
7	4.9560	32.80	9.93	42.73	56.00	-13.27	QP	P	
8 *	4.9560	31.27	9.93	41.20	46.00	-4.80	AVG	P	
9	5.4555	33.80	9.95	43.75	60.00	-16.25	QP	P	
10	5.4555	31.46	9.95	41.41	50.00	-8.59	AVG	P	
11	14.3790	36.33	9.95	46.28	60.00	-13.72	QP	P	
12	14.3790	34.11	9.95	44.06	50.00	-5.94	AVG	P	

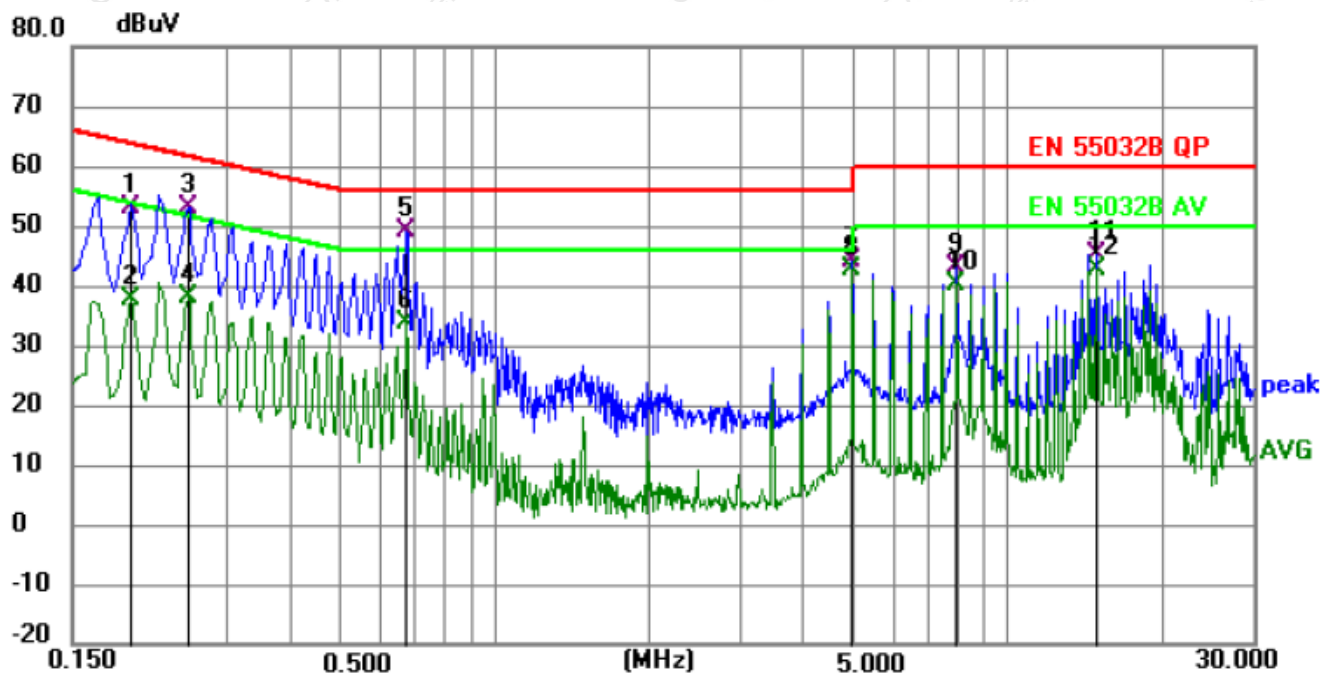
Remark: Correct Factor = Cable lose + LISN insertion loss;

Level = Reading + Correct factor; Margin = Level – Limit;



Conducted Emission Test Data

Temperature:	24.5 °C	Relative Humidity:	54%
Pressure:	1009hPa	Phase:	Neutral
Test Voltage:	AC 230V/50Hz	Test Mode:	Mode 1



No.	Frequency (MHz)	Reading (dBuV)	Factor (dB)	Level (dBuV)	Limit (dBuV)	Margin (dB)	Detector	P/F	Remark
1	0.1949	43.25	9.85	53.10	63.83	-10.73	QP	P	
2	0.1949	27.77	9.85	37.62	53.83	-16.21	AVG	P	
3	0.2535	43.11	9.97	53.08	61.64	-8.56	QP	P	
4	0.2535	27.96	9.97	37.93	51.64	-13.71	AVG	P	
5	0.6720	39.07	9.95	49.02	56.00	-6.98	QP	P	
6	0.6720	23.82	9.95	33.77	46.00	-12.23	AVG	P	
7	4.9694	34.06	9.93	43.99	56.00	-12.01	QP	P	
8 *	4.9694	32.77	9.93	42.70	46.00	-3.30	AVG	P	
9	7.9485	32.90	9.98	42.88	60.00	-17.12	QP	P	
10	7.9485	30.27	9.98	40.25	50.00	-9.75	AVG	P	
11	14.9055	35.28	9.96	45.24	60.00	-14.76	QP	P	
12	14.9055	32.75	9.96	42.71	50.00	-7.29	AVG	P	

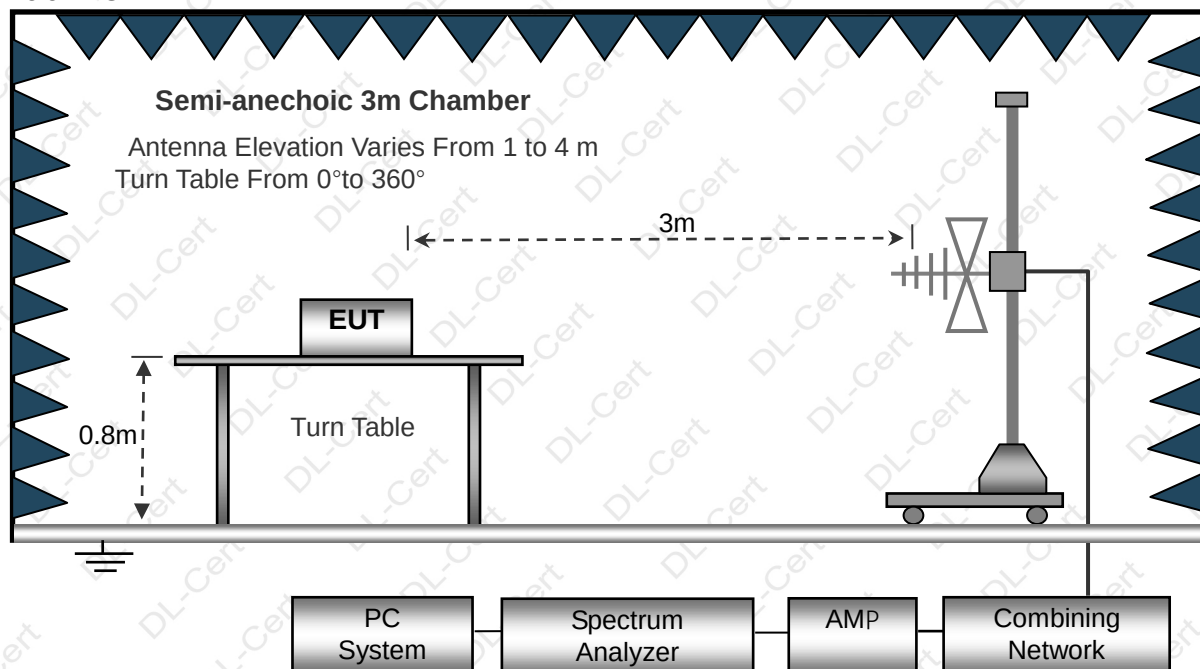
Remark: Correct Factor = Cable lose + LISN insertion loss;

Level = Reading + Correct factor; Margin = Level – Limit;

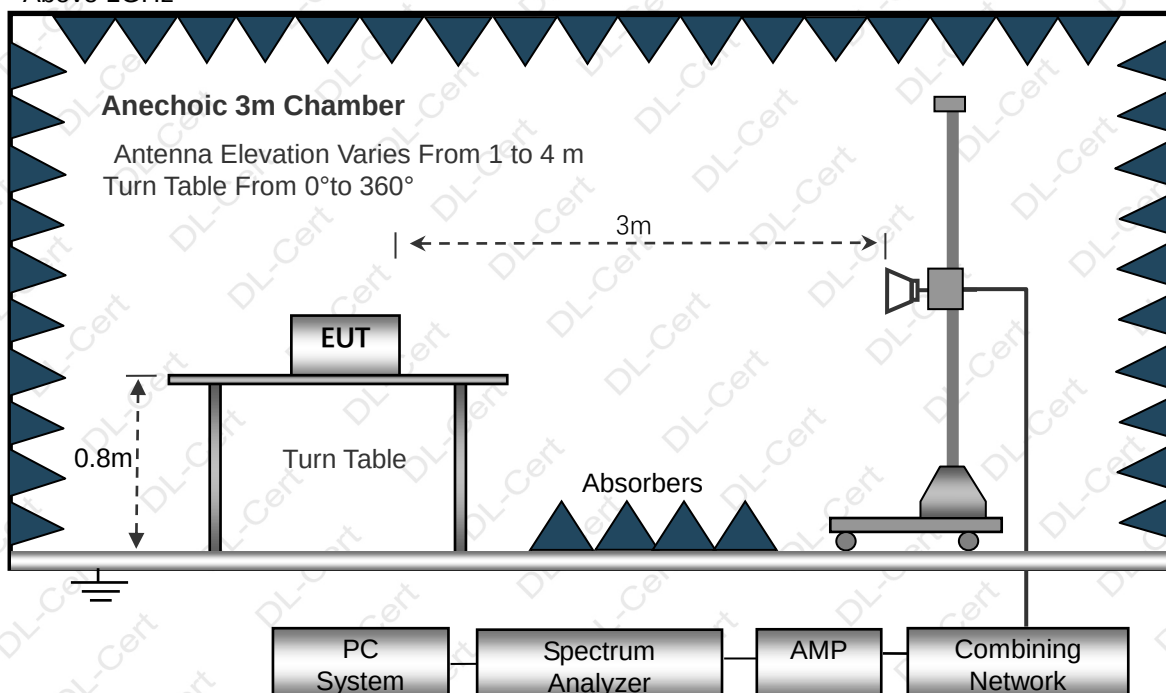
6. RADIATION EMISSION TEST

6.1 Block Diagram of Test Setup

Below 1GHz



Above 1GHz



6.2 Test Standard and Limit

EN 55032



Below 1GHz

Equipment Type	Distance (Meters)	Frequency MHz	Limit values dB(μV/m) Quasi-peak
FM receivers	3	≤1 000	Fundamental 60
		30 to 230	Harmonics 52
		230 to 300	Harmonics 52
		300 to 1 000	Harmonics 56
Other		30 to 230	40
		230 to 1 000	47

Above 1GHz

Frequency MHz	Distance (Meters)	Field Strengths Limits dB(μ V)/m	Detector
1000~6000	3	74.0	PEAK
1000~6000	3	54.0	AVERAGE

Remark:

- (1) The smaller limit shall apply at the cross point between two frequency bands.
- (2) Distance refers to the distance in meters between the measuring instrument, antenna and the closed point of any part of the device or system.

6.3 EUT Configuration on Test

The EN 55032 regulations test method must be used to find the maximum emission during radiated emission test.

The configuration of EUT is the same as used in conducted emission test.

Please refer to Section 5.3.

6.4 Operating Condition of EUT

Same as conducted emission test, which is listed in Section 5.4 except the test set up replaced as Section 6.2.

6.5 Test Procedure

- 1) The radiated emissions test was conducted in a semi-anechoic chamber.
- 2) The tabletop EUT was placed upon a non-metallic table 0.8m above the ground reference plane. And for floor-standing arrangement, the EUT was placed on the horizontal ground reference plane, but separated from metallic contact with the ground reference plane by 0.1m of insulation.
- 3) Before final measurements of radiated emissions, a pre-scan was performed in the spectrum mode with the peak detector to find out the maximum emissions spectrum plots of the EUT.
- 4) The frequencies of maximum emission were determined in the final radiated emissions measurement. At each frequency, the EUT was rotated 360°, and the antenna was raised and lowered from 1 to 4 meters in order to determine the maximum disturbance. Measurements were performed for both horizontal and vertical antenna polarization.
- 5) The bandwidth setting on the field strength meter (R&S Test Receiver ESCI) is set at 120KHz.
- 6) The frequency range from 30MHz to 1000MHz is checked.
- 7) For above 1GHz, the peak emission below the average's limit, so the average's result no recording.

6.6 Test Result

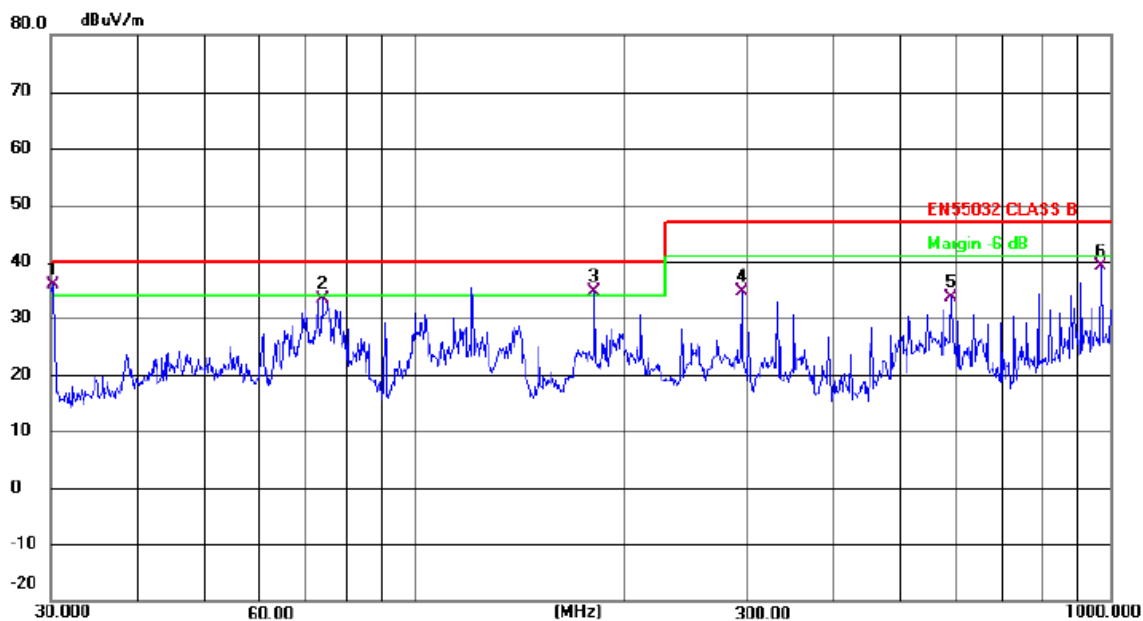
PASS

Please refer to the following page.



Radiation Emission Test Data (Below 1GHz)

Temperature:	24.5 °C	Relative Humidity:	53%
Pressure:	1009hPa	Polarization:	Horizontal
Test Voltage:	AC 230V/50Hz	Test Mode:	Mode 1



No.	Mk.	Freq. MHz	Reading Level dBuV	Correct Factor dB/m	Measure- ment dBuV/m	Limit dBuV/m	Margin dB	Detector	Comment
1	*	30.2111	45.64	-9.85	35.79	40.00	-4.21	QP	
2		73.6170	45.43	-12.11	33.32	40.00	-6.68	QP	
3	!	181.2834	45.02	-10.32	34.70	40.00	-5.30	QP	
4		295.1468	40.50	-5.96	34.54	47.00	-12.46	QP	
5		590.9737	33.13	0.60	33.73	47.00	-13.27	QP	
6		968.9337	32.99	6.24	39.23	47.00	-7.77	QP	

Remark:

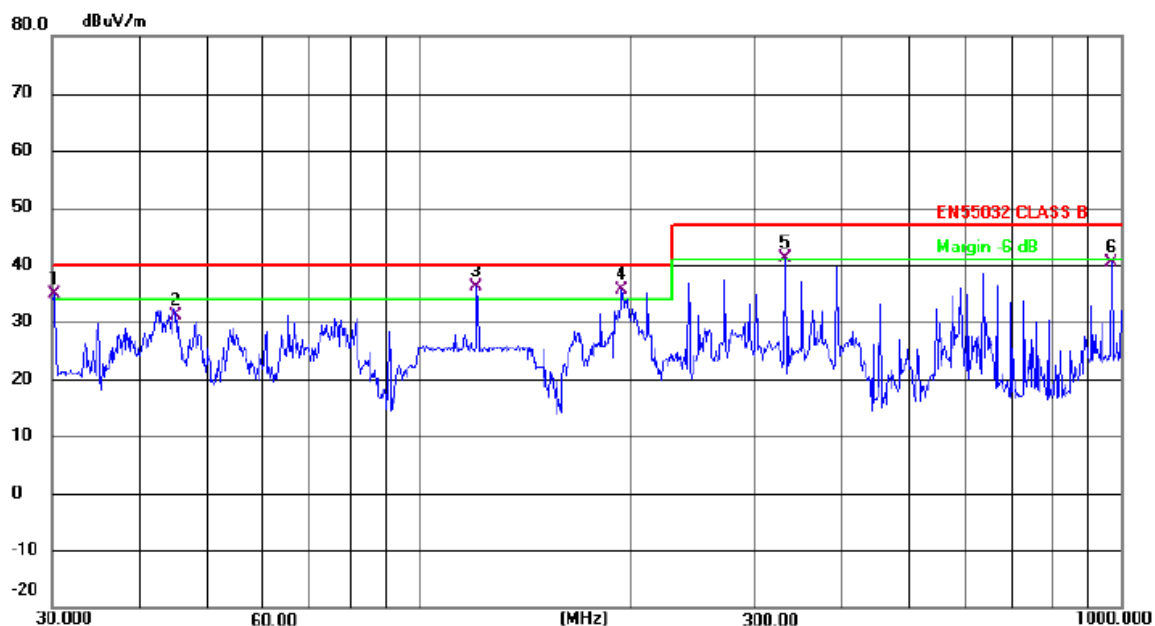
Correct Factor=Cable loss+Antenna factor-Preamplifier

MeasurementLevel= Reading Level + Correct Factor, Margin = Measurement Level- Limit.



Radiation Emission Test Data (Below 1GHz)

Temperature:	24.5 °C	Relative Humidity:	53%
Pressure:	1009hPa	Polarization:	Vertical
Test Voltage:	AC 230V/50Hz	Test Mode:	Mode 1



No.	Mk.	Freq. MHz	Reading Level dBuV	Correct Factor dB/m	Measure- ment dBuV/m	Limit dBuV/m	Margin dB	Detector	Comment
1	!	30.2111	44.62	-9.85	34.77	40.00	-5.23	QP	
2		44.9006	38.05	-6.97	31.08	40.00	-8.92	QP	
3	*	120.6991	46.97	-10.79	36.18	40.00	-3.82	QP	
4	!	194.4534	44.96	-9.36	35.60	40.00	-4.40	QP	
5	!	332.5187	46.51	-5.46	41.05	47.00	-5.95	QP	
6		968.9337	34.26	6.24	40.50	47.00	-6.50	QP	

Remark:

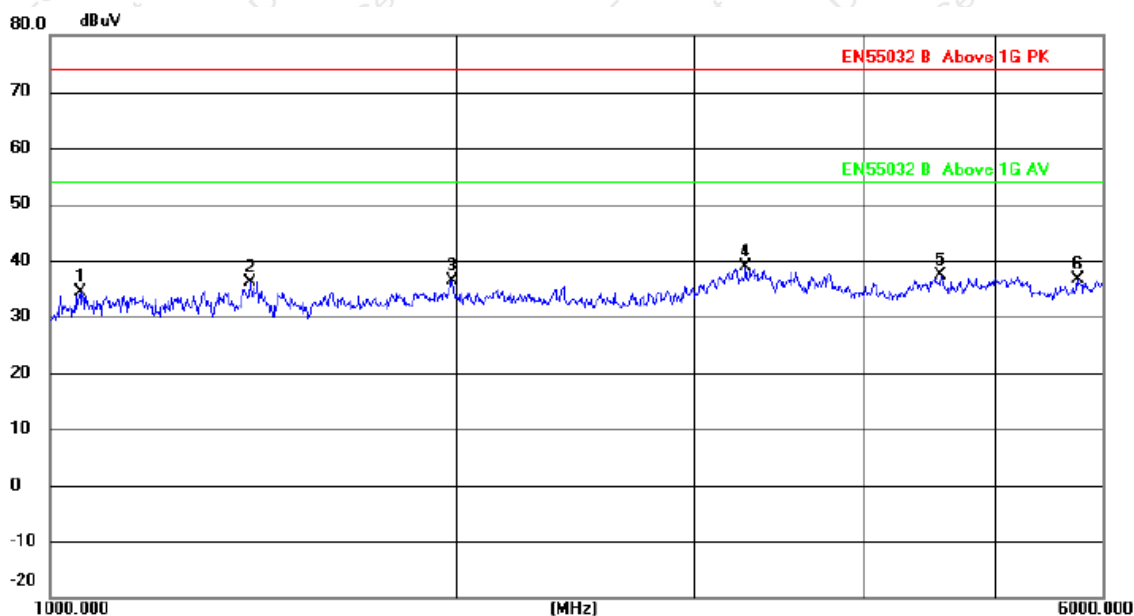
Correct Factor=Cable loss+Antenna factor-Preamplifier

MeasurementLevel= Reading Level + Correct Factor, Margin = Measurement Level- Limit.



Radiation Emission Test Data (Above 1GHz)

Temperature:	24.5 °C	Relative Humidity:	53%
Pressure:	1009hPa	Polarization:	Horizontal
Test Voltage:	AC 230V/50Hz	Test Mode:	Mode 1



No. Mk.	Freq. MHz	Reading Level dBμV	Correct Factor dB	Measure- ment dBμV	Limit dBμV	Margin dB	Detector	Comment
1	1051.449	55.34	-20.85	34.49	74.00	-39.51	peak	
2	1405.558	55.20	-19.05	36.15	74.00	-37.85	peak	
3	1982.685	52.69	-16.39	36.30	74.00	-37.70	peak	
4 *	3268.571	51.16	-12.34	38.82	74.00	-35.18	peak	
5	4561.357	43.53	-6.25	37.28	74.00	-36.72	peak	
6	5747.456	38.95	-2.32	36.63	74.00	-37.37	peak	

Remark:

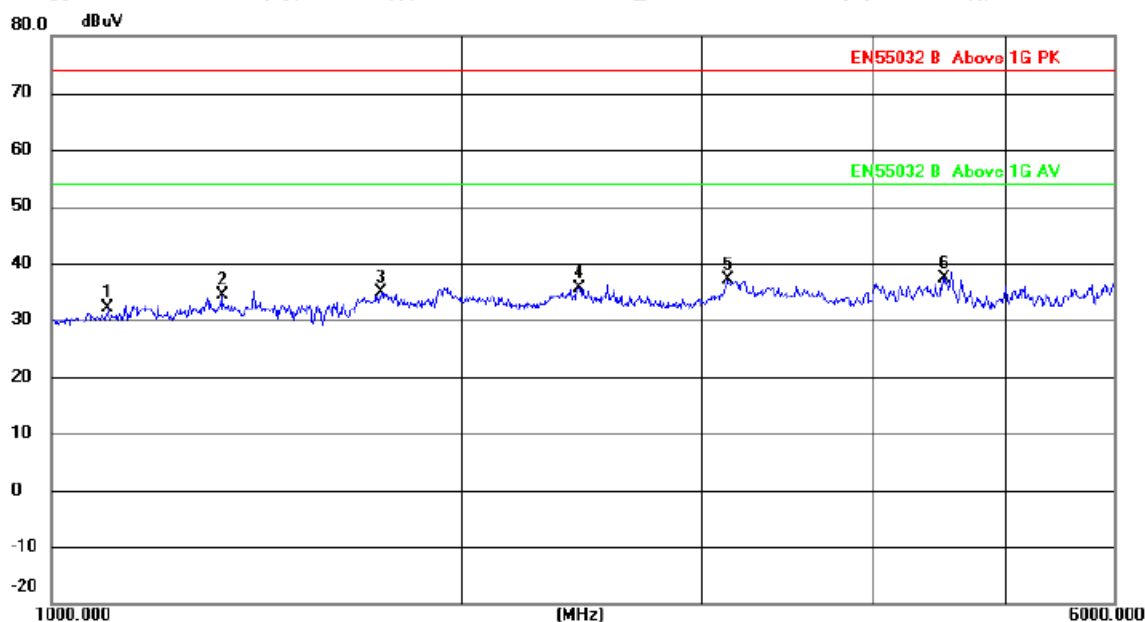
Correct Factor=Cable loss+Antenna factor-Preamplifier

MeasurementLevel= Reading Level + Correct Factor, Margin = Measurement Level- Limit.



Radiation Emission Test Data (Above 1GHz)

Temperature:	24.5 °C	Relative Humidity:	53%
Pressure:	1009hPa	Polarization:	Vertical
Test Voltage:	AC 230V/50Hz	Test Mode:	Mode 1



No.	Mk.	Freq. MHz	Reading Level dBuV	Correct Factor dB	Measure- ment dBuV	Limit dBuV	Margin dB	Detector	Comment
1		1099.618	52.70	-20.61	32.09	74.00	-41.91	peak	
2		1332.000	54.13	-19.66	34.47	74.00	-39.53	peak	
3		1745.842	52.95	-17.97	34.98	74.00	-39.02	peak	
4		2431.997	50.56	-14.85	35.71	74.00	-38.29	peak	
5		3125.390	48.38	-11.37	37.01	74.00	-36.99	peak	
6	*	4512.583	43.98	-6.50	37.48	74.00	-36.52	peak	

Remark:

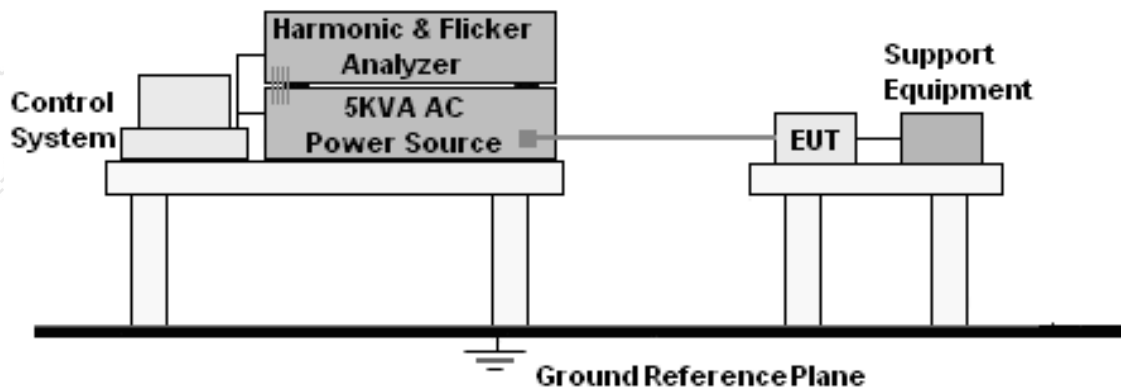
Correct Factor=Cable loss+Antenna factor-Preamplifier

MeasurementLevel= Reading Level + Correct Factor, Margin = Measurement Level- Limit.



7. HARMONIC CURRENT EMISSION TEST

7.1 Block Diagram of Test Setup



7.2 Test Standard

EN IEC 61000-3-2

7.3 Operating Condition of EUT

Setup the EUT as shown in Section 5.1.

Turn on the power of all equipment.

Let the EUT work in test mode and test it.

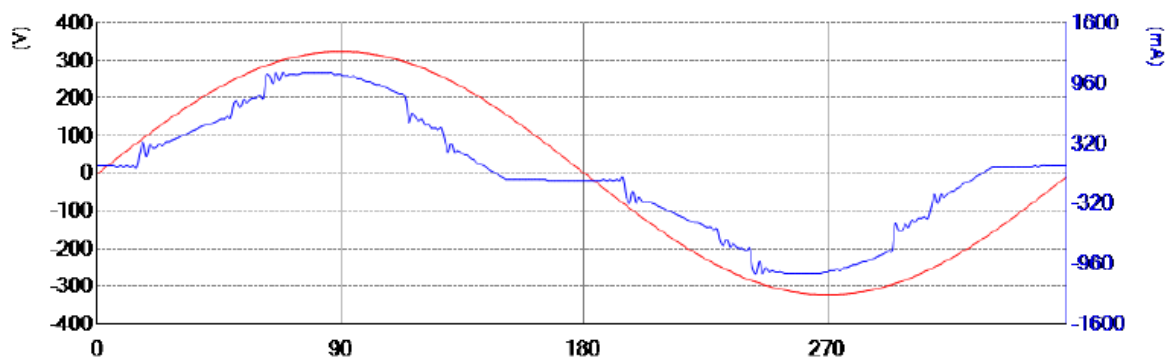
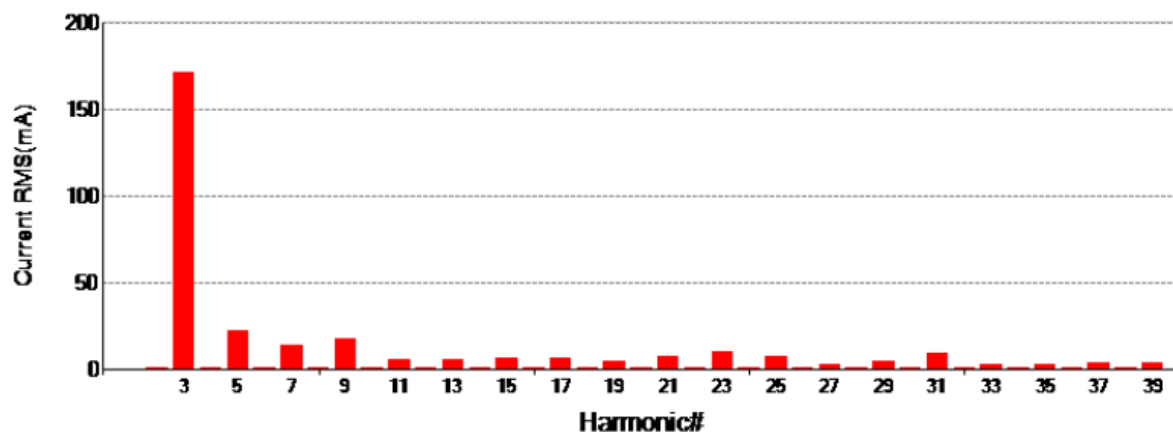
7.4 Test Procedure

The power cord of the EUT is connected to the output of the test system. Turn on the power of the EUT and use the test system to test the harmonic current level.

7.5 Test Results

PASS

Please refer to the following page.

Test Result: **pass**Source qualification(Power Off Load): **Idle - Pass****Current & voltage waveforms****Waveform Graph****Harmonics and Class A****Harmonics Histogram**



THC(mA): 176.200

I - THD(%): 29.6

POHC(mA):19.500

POHC Limit(mA):251.353

Parameter values during test:

V_RMS (Volts): 229.8

I_RMS(mA): 594.4

Power (Watts): 127.5

Frequency(Hz): 50.0

Crest Factor: 1.663

Power Factor: 0.933

Harm#	Harms(filtered) (mA)	Limit (mA)	Harms(avg) (mA)	100%Limit	Harms(max) (mA)	150%Limit	Status
I_Fund	567.000						
2	1.000	1080.000	0.900	0.083	1.500	0.093	Pass
3	171.600	2300.000	171.500	7.457	171.700	4.977	Pass
4	0.400	430.000	0.500	0.116	0.600	0.093	Pass
5	22.400	1140.000	22.500	1.974	22.600	1.322	Pass
6	0.500	300.000	0.500	0.167	0.700	0.156	Pass
7	14.100	770.000	14.000	1.818	14.100	1.221	Pass
8	0.500	230.000	0.500	0.217	0.600	0.174	Pass
9	17.400	400.000	17.600	4.400	17.800	2.967	Pass
10	0.400	184.000	0.500	0.272	0.500	0.181	Pass
11	5.500	330.000	5.600	1.697	5.600	1.131	Pass
12	0.400	153.300	0.500	0.326	0.500	0.217	Pass
13	5.700	210.000	5.700	2.714	5.700	1.810	Pass
14	0.400	131.400	0.500	0.381	0.500	0.254	Pass
15	6.200	150.000	6.200	4.133	6.300	2.800	Pass
16	0.400	115.000	0.500	0.435	0.500	0.290	Pass
17	6.800	132.400	6.900	5.211	7.000	3.525	Pass
18	0.400	102.200	0.500	0.489	0.500	0.326	Pass
19	4.700	118.400	4.800	4.054	4.900	2.759	Pass
20	0.400	92.000	0.500	0.543	0.500	0.362	Pass
21	7.000	107.100	7.300	6.816	7.400	4.606	Pass
22	0.500	83.600	0.600	0.718	0.600	0.478	Pass
23	10.100	97.800	10.000	10.225	10.100	6.885	Pass
24	0.600	76.700	0.600	0.782	0.600	0.522	Pass
25	7.600	90.000	7.600	8.444	7.700	5.704	Pass
26	0.500	70.800	0.500	0.706	0.600	0.565	Pass
27	3.200	83.300	3.300	3.962	3.500	2.801	Pass
28	0.400	65.700	0.500	0.761	0.600	0.609	Pass
29	5.000	77.600	4.900	6.314	5.100	4.381	Pass
30	0.500	61.300	0.500	0.816	0.600	0.653	Pass
31	9.100	72.600	9.200	12.672	9.300	8.540	Pass
32	0.500	57.500	0.600	1.043	0.700	0.812	Pass
33	2.600	68.200	2.500	3.666	2.600	2.542	Pass
34	0.500	54.100	0.500	0.924	0.600	0.739	Pass
35	3.200	64.300	3.300	5.132	3.400	3.525	Pass
36	0.500	51.100	0.500	0.978	0.600	0.783	Pass
37	3.400	60.800	3.600	5.921	3.700	4.057	Pass
38	0.400	48.400	0.500	1.033	0.600	0.826	Pass
39	4.100	57.700	4.200	7.279	4.300	4.968	Pass
40	0.500	46.000	0.600	1.304	0.600	0.870	Pass

Note: All harmonics are below the minimum limits and are ignored.



8. VOLTAGE FLUCTUATIONS & FLICKER TEST

8.1 Block Diagram of Test Setup

Same as Section 7.1.

8.2 Test Standard

EN 61000-3-3

8.3 Operating Condition of EUT

Same as Section 7.3. The power cord of the EUT is connected to the output of the test system. Turn on the power of the EUT and use the test system to test the harmonic current level.

Flicker Test Limit

Test items	Limits
Pst	1.0
dc	3.3%
Tmax	4.0%
dt	Not exceed 3.3% for 500ms

8.4 Test Procedure

The power cord of the EUT is connected to the output of the test system. Turn on the power of the EUT and use the test system to test the harmonic current level.

8.5 Test Results

PASS

Flicker Test Data			
Temperature:	24.5 °C	Relative Humidity:	54%
Test Voltage:	AC 230V/50Hz	Test Mode:	Mode 1

Voltage Fluctuation	Limit	Value
Relative Voltage Change Characteristic Tmax (dc > 3%)	500ms	0ms
Maximum Relative Voltage Change dmax	4%	0.00
	6%	/
	7%	/
Relative Steady-state Voltage Change dc	3.3%	0.00

Flicker	Limit	Value
Short-term Flicker Indicator Pst	1.0	0.063
Long-term Flicker Indicator Plt	0.65	/

**9. IMMUNITY TEST OF GENERAL THE PERFORMANCE CRITERIA**

Product Standard	EN 55035, ETSI EN 301 489-17	
Criteria	During the test	After the test
A	Shall operate as intended. (see note). Shall be no loss of function. Shall be no unintentional transmissions.	Shall operate as intended. Shall be no degradation of performance. Shall be no loss of function. Shall be no loss of critical stored data.
B	May be loss of function.	Functions shall be self-recoverable. Shall operate as intended after recovering. Shall be no loss of critical stored data.
C	May be loss of function.	Functions shall be recoverable by the operator. Shall operate as intended after recovering. Shall be no loss of critical stored data.
NOTE: Operate as intended during the test allows a level of degradation in accordance with clause 6.2.2. 6.2.2 Minimum performance level For equipment that supports a PER or FER, the minimum performance level shall be a PER or FER less than or equal to 10 %. For equipment that does not support a PER or a FER, the minimum performance level shall be no loss of the wireless transmission function needed for the intended use of the equipment.		

**PERFORMANCE FOR TT**

The performance criteria B shall apply, except for voltage dips of 100 ms and voltage interruptions of 5 000 ms duration, for which performance criteria C shall apply.

Tests shall be repeated with the EUT in standby mode (if applicable) to ensure that unintentional transmission does not occur. In systems using acknowledgement signals, it is recognized that an acknowledgement (ACK) or not-acknowledgement (NACK) transmission may occur, and steps should be taken to ensure that any transmission resulting from the application of the test is correctly interpreted.

PERFORMANCE FOR TR

The performance criteria B shall apply, except for voltage dips of 100ms and voltage interruptions of 5 000ms duration for which performance criteria C shall apply.

Where the EUT is a transceiver, under no circumstances, shall the transmitter operate unintentionally during the test. In systems using acknowledgement signals, it is recognized that an ACK or NACK transmission may occur, and steps should be taken to ensure that any transmission resulting from the application of the test is correctly interpreted.

PERFORMANCE FOR CT

The performance criteria A shall apply.

Tests shall be repeated with the EUT in standby mode (if applicable) to ensure that unintentional transmission does not occur. In systems using acknowledgement signals, it is recognized that an ACKnowledgement (ACK) or Not ACKnowledgement (NACK) transmission may occur, and steps should be taken to ensure that any transmission resulting from the application of the test is correctly interpreted.

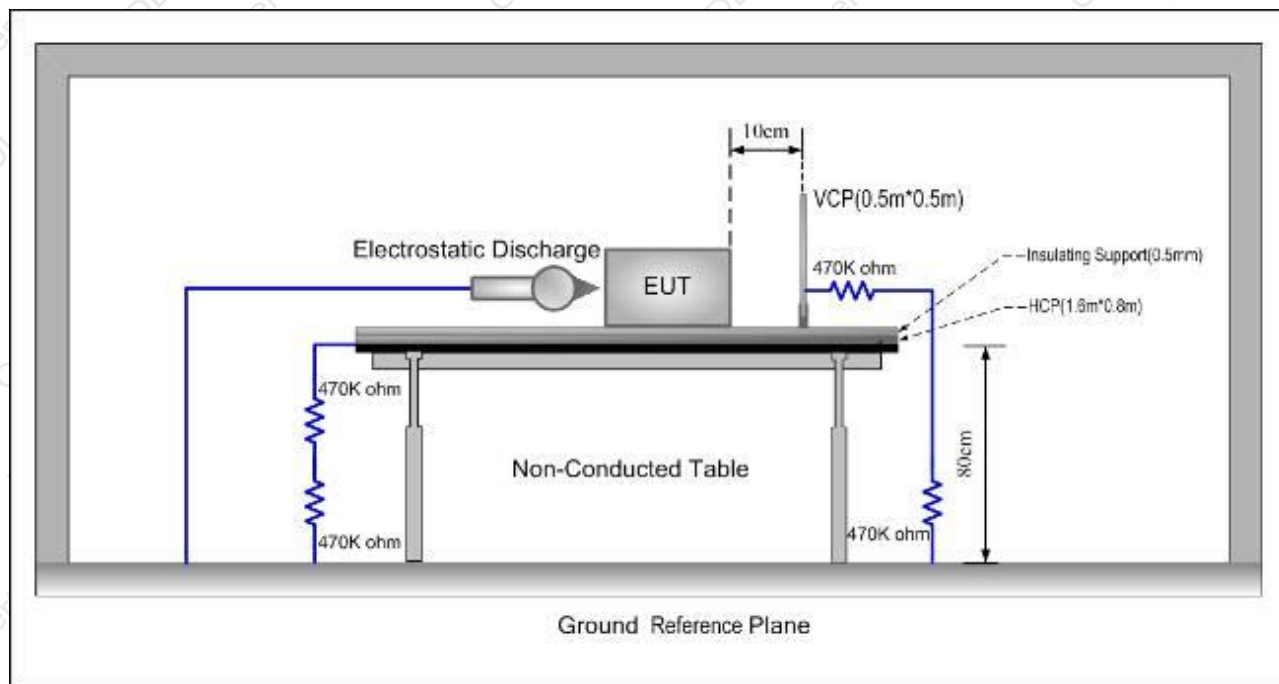
PERFORMANCE FOR CR

The performance criteria A shall apply.

Where the EUT is a transceiver, under no circumstances, shall the transmitter operate unintentionally during the test. In systems using acknowledgement signals, it is recognized that an ACK or NACK transmission may occur, and steps should be taken to ensure that any transmission resulting from the application of the test is correctly interpreted.

10. ELECTROSTATIC DISCHARGE IMMUNITY TEST

10.1 Block Diagram of Test Setup



10.2 Test Standard

ETSI EN 301 489-17, EN 55035, EN 61000-4-2

10.3 Severity Levels and Performance Criterion

Severity Level: 3 / Air Discharge: $\pm 8\text{KV}$

Level: 2 / Contact Discharge: $\pm 4\text{KV}$

Performance criterion: B

10.4 Test Procedure

- Electrostatic discharges were applied only to those points and surfaces of the Product that are accessible to users during normal operation.
- The test was performed with at least ten single discharges on the pre-selected points in the most sensitive polarity.
- The time interval between two successive single discharges was at least 1 second.
- The ESD generator was held perpendicularly to the surface to which the discharge was applied and the return cable was at least 0.2 meters from the Product.
- Contact discharges were applied to the non-insulating coating, with the pointed tip of the generator penetrating the coating and contacting the conducting substrate.
- Air discharges were applied with the round discharge tip of the discharge electrode approaching the Product as fast as possible (without causing mechanical damage) to touch the Product. After each discharge, the ESD generator was removed from the Product and re-triggered for a new single discharge. The test was repeated until all discharges were complete.
- At least ten single discharges (in the most sensitive polarity) were applied to the Horizontal Coupling Plane at points on each side of the Product. The ESD generator was positioned vertically at a distance of 0.1 meters from the Product with the discharge electrode touching the HCP.



- h. At least ten single discharges (in the most sensitive polarity) were applied to the center of one vertical edge of the Vertical Coupling Plane in sufficiently different positions that the four faces of the Product were completely illuminated. The VCP (dimensions 0.5m x 0.5m) was placed vertically to and 0.1 meters from the Product.

10.5 Test Results

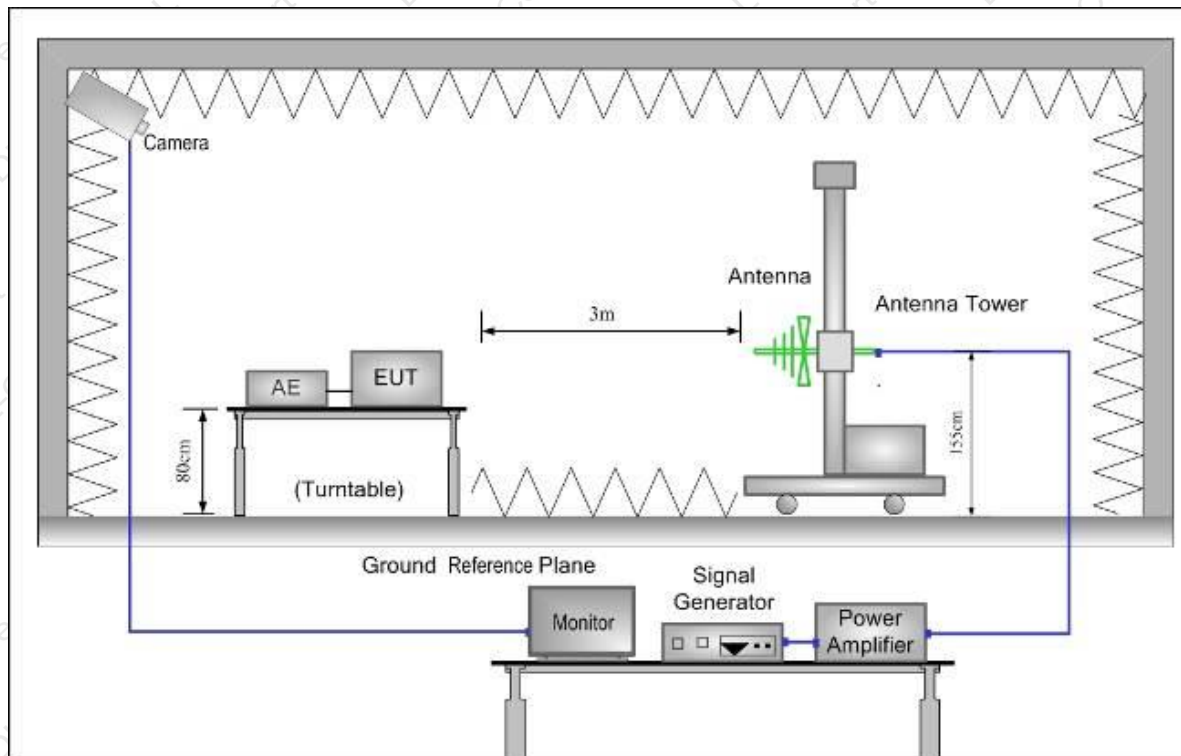
PASS

Electrostatic Discharge Test Data					
Temperature:	25.1℃	Humidity:	55%		
Power Supply:	AC 230V/50Hz	Test Mode:	Mode 1		
Discharge Method	Discharge Position	Voltage (±kV)	Min. No. of Discharge per polarity (Each Point)	Required Level	Result
Contact Discharge	Conductive Surfaces	4	10	B	Pass
	Indirect Discharge HCP	4	10	B	Pass
	Indirect Discharge VCP	4	10	B	Pass
Air Discharge	Slots, Apertures, and Insulating Surfaces	8	10	B	Pass
Note: N/A					



11. RF FIELD STRENGTH SUSCEPTIBILITY TEST

11.1 Block Diagram of Test Setup



11.2 Test Standard

ETSI EN 301 489-17, EN 55035, EN IEC 61000-4-3

11.3 Severity Levels and Performance Criterion

Severity Level 2, 3V / m

Performance criterion: A

11.4 Test Procedure

The EUT and its simulators are placed on a turn table which is 0.8 meter above ground. EUT is set 3 meter away from the transmitting antenna which is mounted on an antenna tower. Both horizontal and vertical polarization of the antenna are set on test. Each of the four sides of EUT must be faced this transmitting antenna and measured individually.

All the scanning conditions are as follows:

Condition of Test	Remarks
Fielded Strength	3 V/m (Severity Level 2)
Radiated Signal	Modulated
Scanning Frequency	80 – 6000 MHz
Dwell time of radiated	0.0015 decade/s
Waiting Time	1 Sec.



11.5 Test Results

PASS

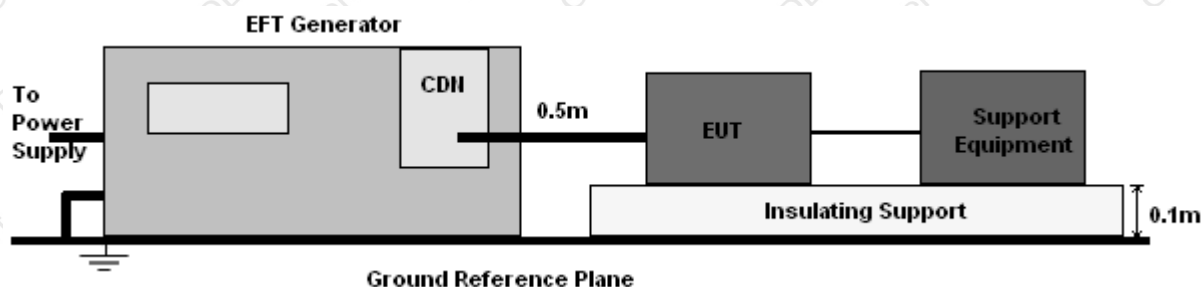
R/S Test Data				
Temperature:	25.1℃		Humidity:	55%
Power Supply:	AC 230V/50Hz		Test Mode:	Mode 1
Criterion:	A		Steps	1 %
Frequency (MHz)	Position	Field Strength (V/m)	Required Level	Result
80 – 6000	Front, Right, Back, Left	3	A	Pass
Note: N/A				



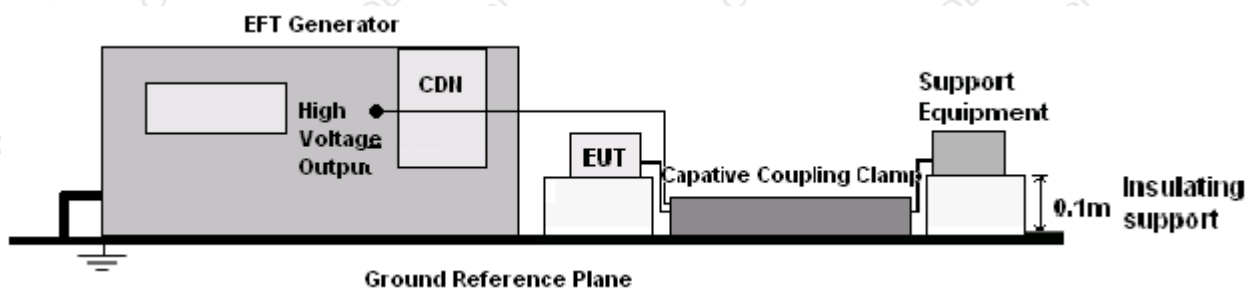
12. ELECTRICAL FAST TRANSIENT/BURST IMMUNITY TEST

12.1 Block Diagram of EUT Test Setup

For input a.c. / d.c. power port:



For signal lines and control lines:



12.2 Test Standard

ETSI EN 301 489-17, EN 55035, EN 61000-4-4

12.3 Severity Levels and Performance Criterion

Severity Level 2 at 1KV, Pulse Rise time & Duration: 5 nS / 50 nS

Performance criterion: B

12.4 Test Procedure

EUT shall be placed 0.8m high above the ground reference plane which is a min.1m*1m metallic sheet with 0.65mm minimum thickness. This reference ground plane shall project beyond the EUT by at least 0.1m on all sides and the minimum distance between EUT and all other conductive structure, except the ground plane beneath the EUT, shall be more than 0.5m

For input and output AC power ports:

The EUT is connected to the power mains by using a coupling device which couples the EFT interference signal to AC power lines. Both polarities of the test voltage should be applied during compliance test and the duration of the test is 2 minutes.



12.5 Test Results

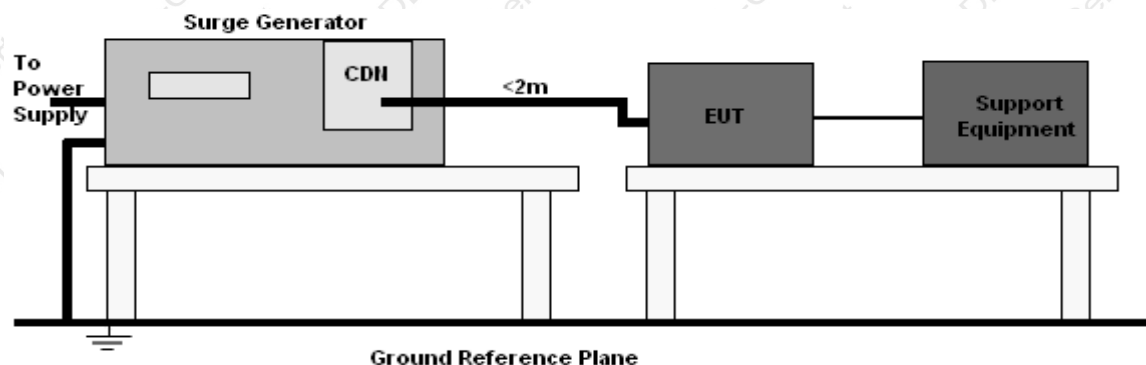
PASS

EFT Test Data			
Temperature:	24.5°C	Humidity:	53%
Power Supply :	AC 230V/50Hz	Test Mode:	Mode 1
Coupling Line	Test Voltage(kV)	Performance Criterion	Result
L	±0.5, 1	B	PASS
N	±0.5, 1	B	PASS
L-N	±0.5, 1	B	PASS
PE	±0.5, 1	B	PASS
L-PE	±0.5, 1	B	PASS
N-PE	±0.5, 1	B	PASS
L-N-PE	±0.5, 1	B	PASS
Note: N/A			



13. SURGE TEST

13.1 Block Diagram of EUT Test Setup



13.2 Test Standard

ETSI EN 301 489-17, EN 55035, EN 61000-4-5

13.3 Severity Levels and Performance Criterion

Severity Level: Line to Line, Level 2 at 1KV;

Severity Level: Line to Earth, Level 3 at 2KV.

Performance criterion: B

13.4 Test Procedure

- 1) Set up the EUT and test generator as shown on section 11.1
- 2) For line to line coupling mode, provide a 1KV 1.2/50us voltage surge (at open-circuit condition) and 8/20us current surge to EUT selected points.
- 3) At least 5 positive and 5 negative (polarity) tests with a maximum 1/min repetition rate are conducted during test.
- 4) Different phase angles are done individually.
- 5) Repeat procedure 2) to 4) except the open-circuit test voltage change from 1KV to 2KV for line to earth coupling mode test.
- 6) Record the EUT operating situation during compliance test and decide the EUT immunity criterion for above each test.

13.5 Test Result

PASS

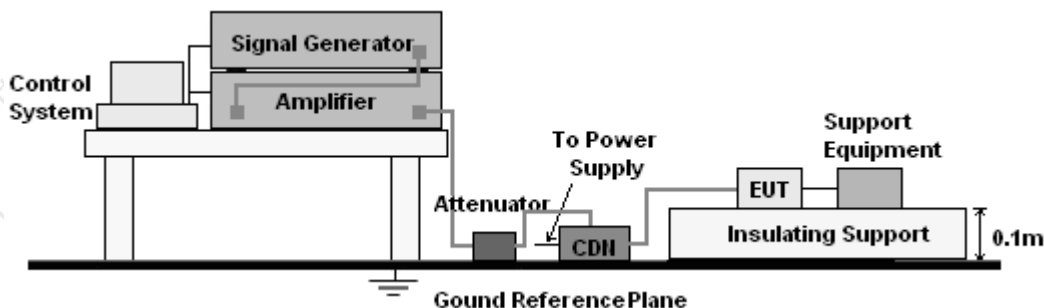
Surge Test Data						
Temperature:		24.5°C		Humidity:		53%
Power Supply :		AC 230V/50Hz		Test Mode:		Mode 1
Location	Polarity	Phase Angle	No of Pulse	Pulse Voltage (KV)	Performance Criterion	Result
L-N	±	0, 90, 180, 270	5	1	B	Pass
L-PE	±	0, 90, 180, 270	5	2	B	Pass
N-PE	±	0, 90, 180, 270	5	2	B	Pass
Note: N/A						



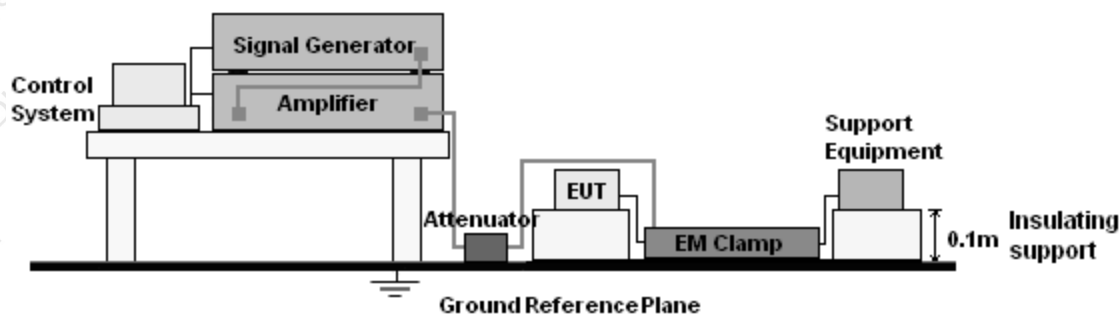
14. INJECTED CURRENTS SUSCEPTIBILITY TEST

14.1 Block Diagram of EUT Test Setup

For input a.c. / d.c. power port:



For signal lines and control lines:



14.2 Test Standard

ETSI EN 301 489-17, EN 55035, EN IEC 61000-4-6

14.3 Severity Levels and Performance Criterion

Severity Level 2: 3V(rms), 150KHz ~ 80MHz

Performance criterion: A

14.4 Test Procedure

- 1) Set up the EUT, CDN and test generator as shown on section 12.1
- 2) Let EUT work in test mode and measure.
- 3) The EUT and supporting equipments are placed on an insulating support 0.1m high above a ground reference plane. CDN (coupling and decoupling device) is placed on the ground plane at above 0.1-0.3m from EUT. Cables between CDN and EUT are as short as possible, and their height above the ground reference plane shall be between 30 and 50 mm (where possible).
- 4) The disturbance signal described below is injected to EUT through CDN.
- 5) The EUT operates within its operational mode(s) under intended climatic conditions after power on.
- 6) The frequency range is swept from 150KHz to 80MHz using 3V signal level, and with the disturbance signal 80% amplitude modulated with a 1KHz sine wave
- 7) The rate of sweep shall not exceed 1.5×10^{-3} decades/s. Where the frequency is swept incrementally, the step size shall not exceed 1% of the start and thereafter 1% of the preceding frequency value.
- 8) Recording the EUT operating situation during compliance test and decide the EUT immunity criterion for above each test.



14.5 Test Result

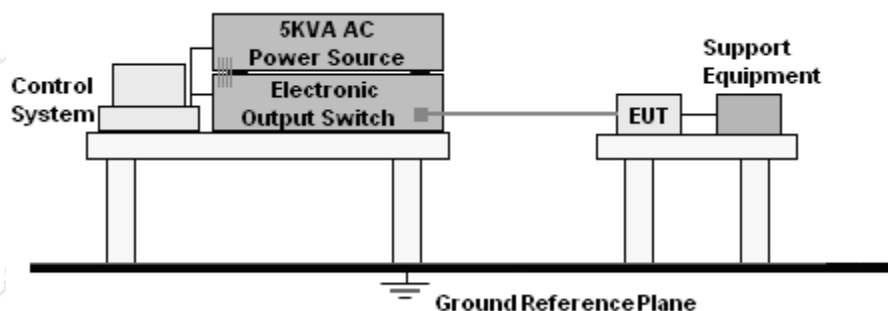
PASS

CS Test Data						
Temperature:		24.5℃		Humidity:		53%
Power Supply :		AC 230V/50Hz		Test Mode:		Mode 1
Frequency Range (MHz)	Injected Position	Strength	Modulation Signal	Freq. Step	Performance Criterion	Result
0.15 ~ 80	AC Line	3V(rms), Unmodulated	AM 80%, 1kHz sine wave	1%	A	Pass
0.15 ~ 80	DC Line, Signal Line	3V(rms), Unmodulated	AM 80%, 1kHz sine wave	1%	/	/
Note: N/A						



15. VOLTAGE DIPS AND INTERRUPTIONS TEST

15.1 Block Diagram of EUT Test Setup



15.2 Test Standard

ETSI EN 301 489-17, EN 55035, EN IEC 61000-4-11

15.3 Severity Levels and Performance Criterion

Input and Output AC Power Ports.

- ☒ Voltage Dips.
- ☒ Voltage Interruptions.

Environmental Phenomena	Test Specification	Units	Performance Criterion
Voltage Dips	100 0.5	% Reduction period	B
	100 1	% Reduction period	B
	30 25	% Reduction period	C
Voltage Interruptions	100 250	% Reduction period	C

15.4 Test Procedure

- 1) Set up the EUT and test generator as shown on section 14.1
- 2) The interruption is introduced at selected phase angles with specified duration. There is a 3mins minimum interval between each test event.
- 3) After each test a full functional check is performed before the next test.
- 4) Repeat procedures 2 & 3 for voltage dips, only the level and duration is changed.
- 5) Record any degradation of performance.



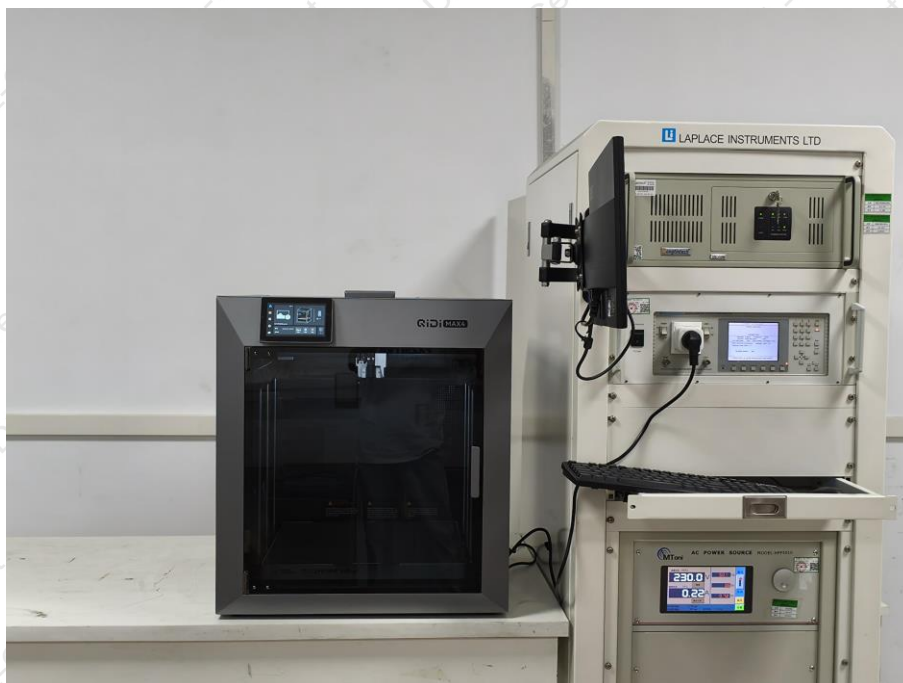
15.5 Test Result
PASS

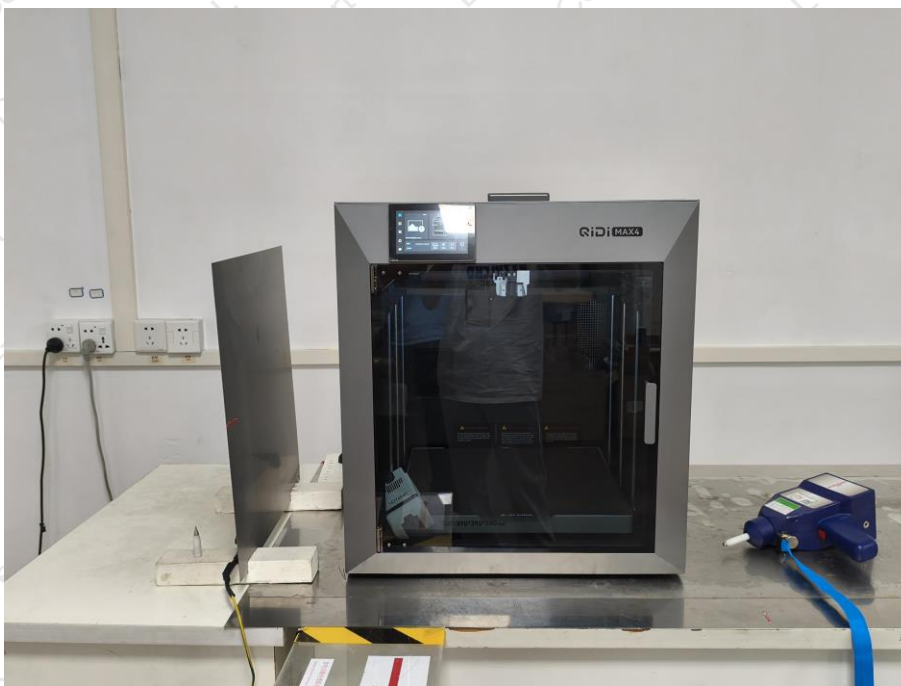
DIPS Test Data				
Temperature:	24.5°C	Humidity:	53%	
Power Supply :	AC 230V/50Hz	Test Mode:	Mode 1	
Environmental Phenomena	Test Specification	Units	Performance Criterion	Result
Voltage Dips	100 0.5	% Reduction period	B	Pass
	100 1	% Reduction period	B	Pass
	30 25	% Reduction period	C	Pass
Voltage Interruptions	100 250	% Reduction period	C	Pass

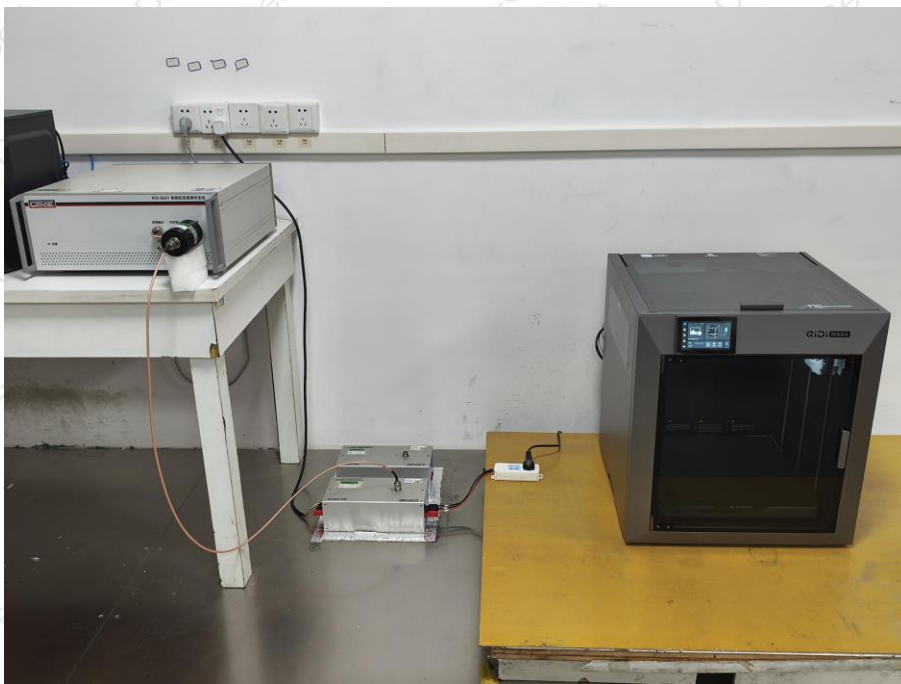


16. SETUP PHOTOGRAPHS



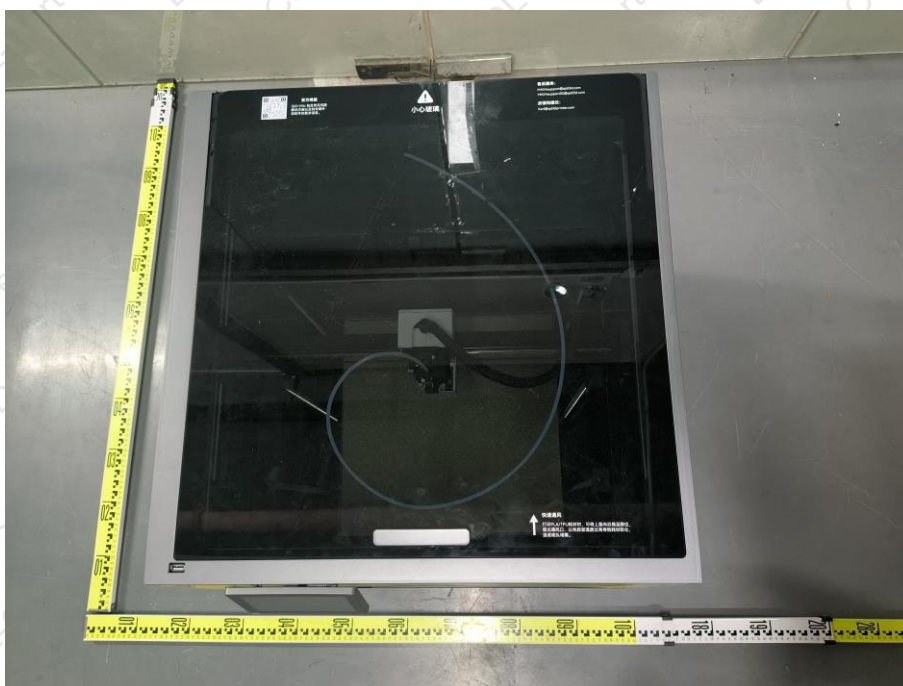
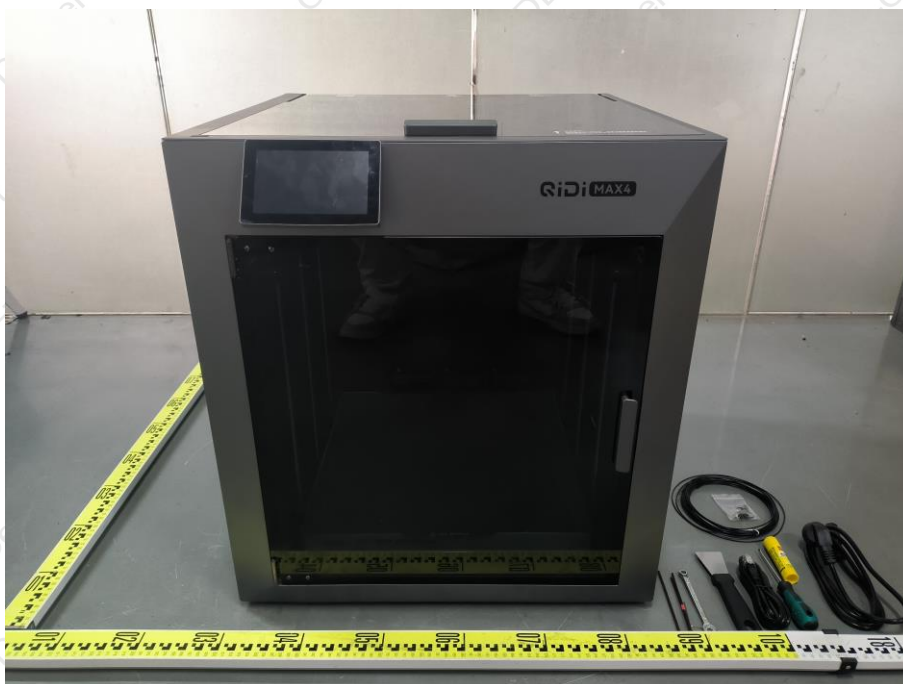




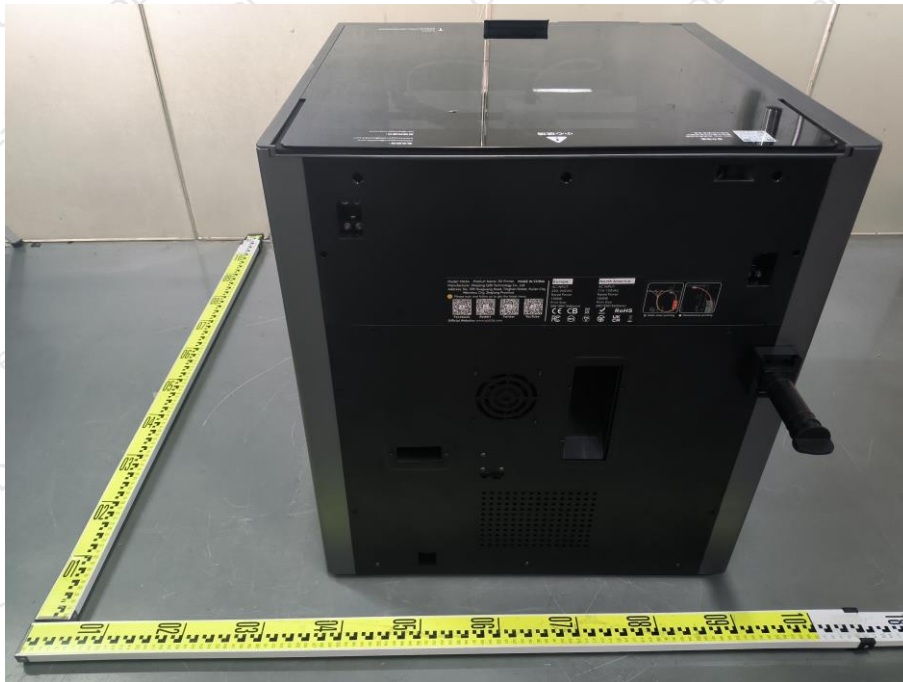
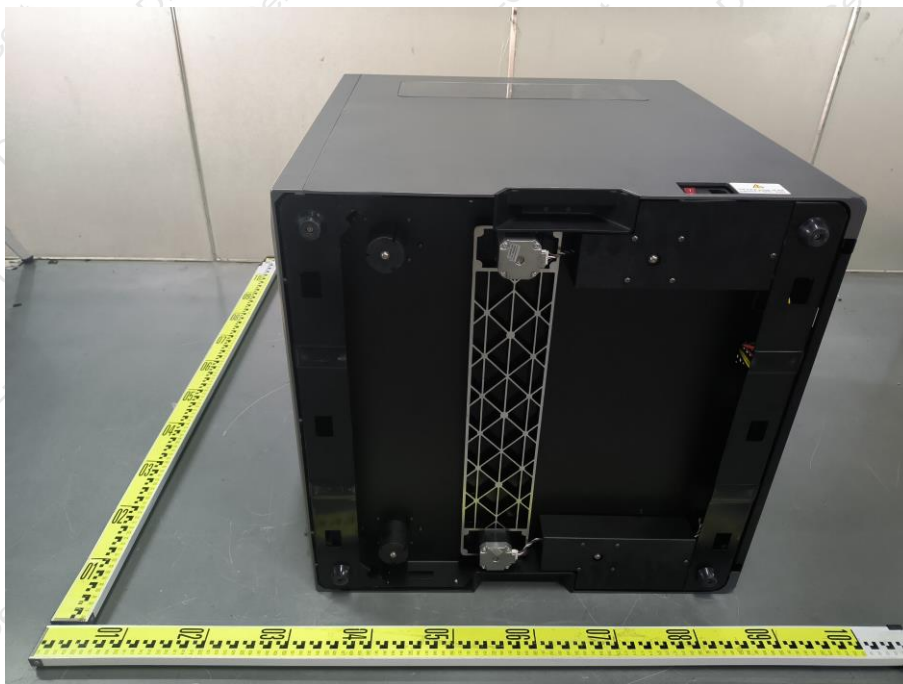




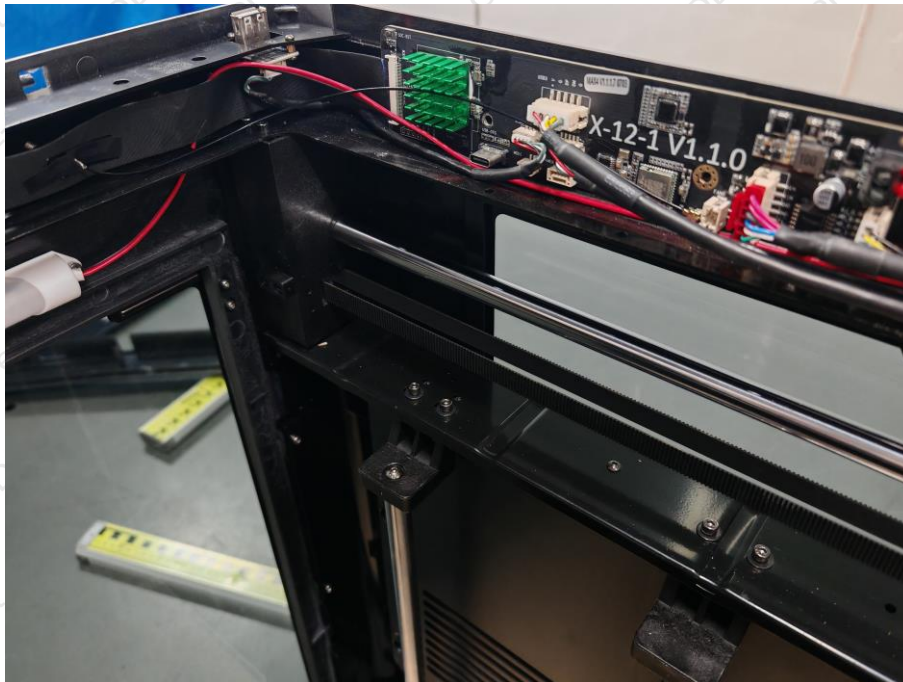
17. EUT PHOTOGRAPHS

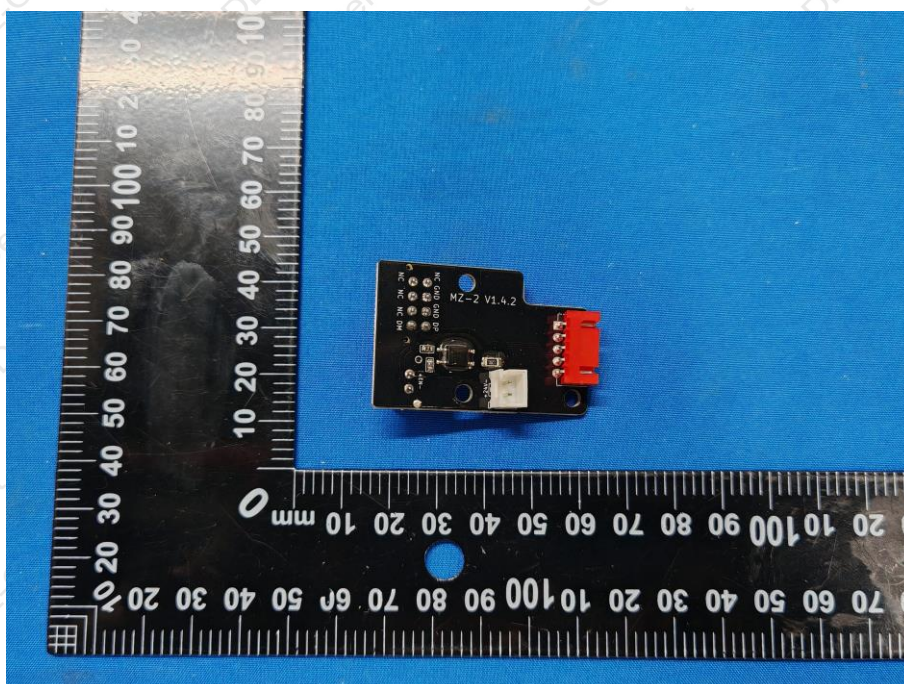
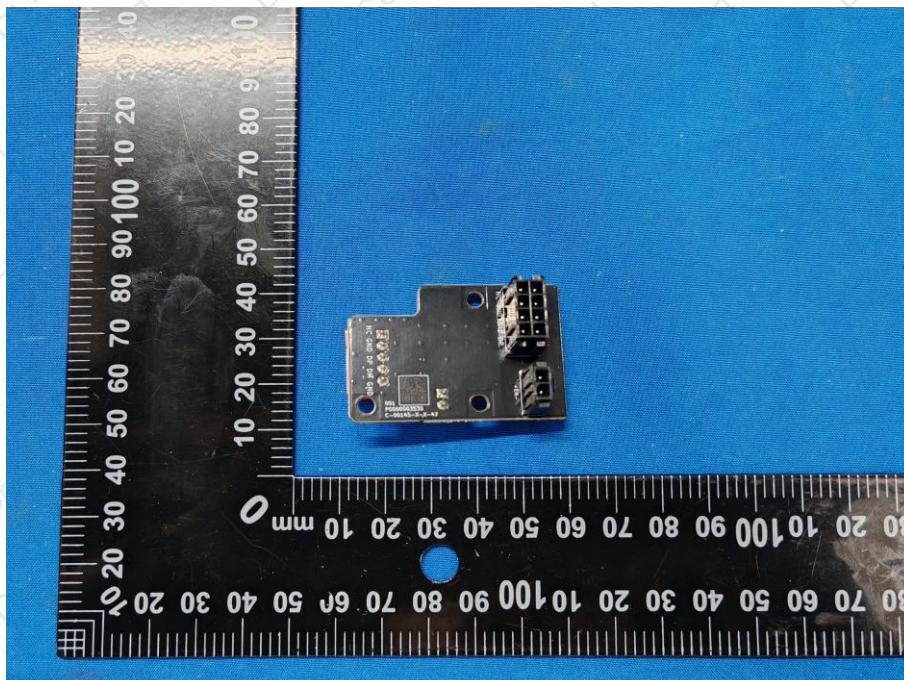


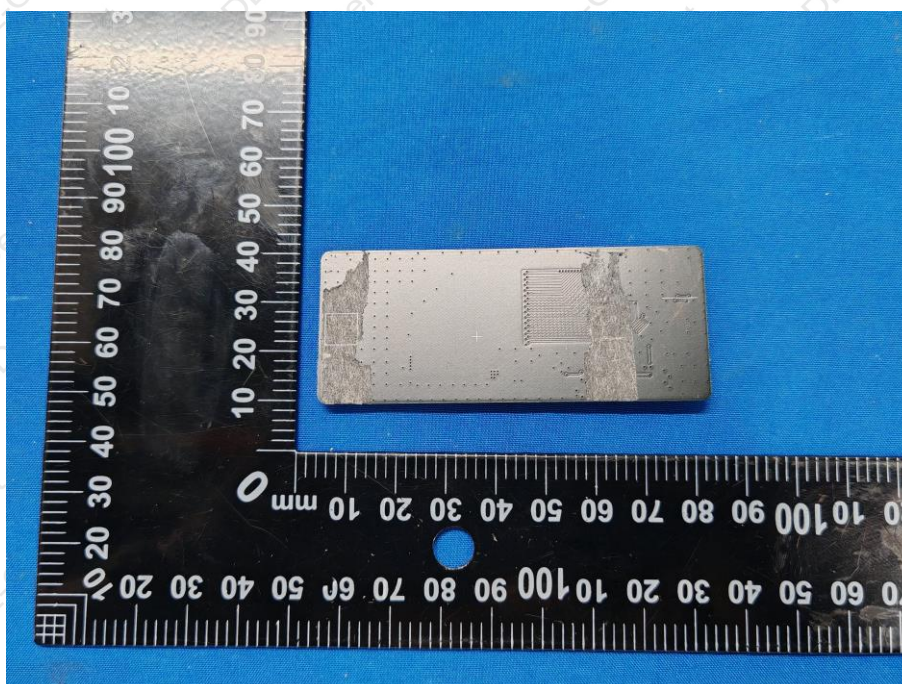
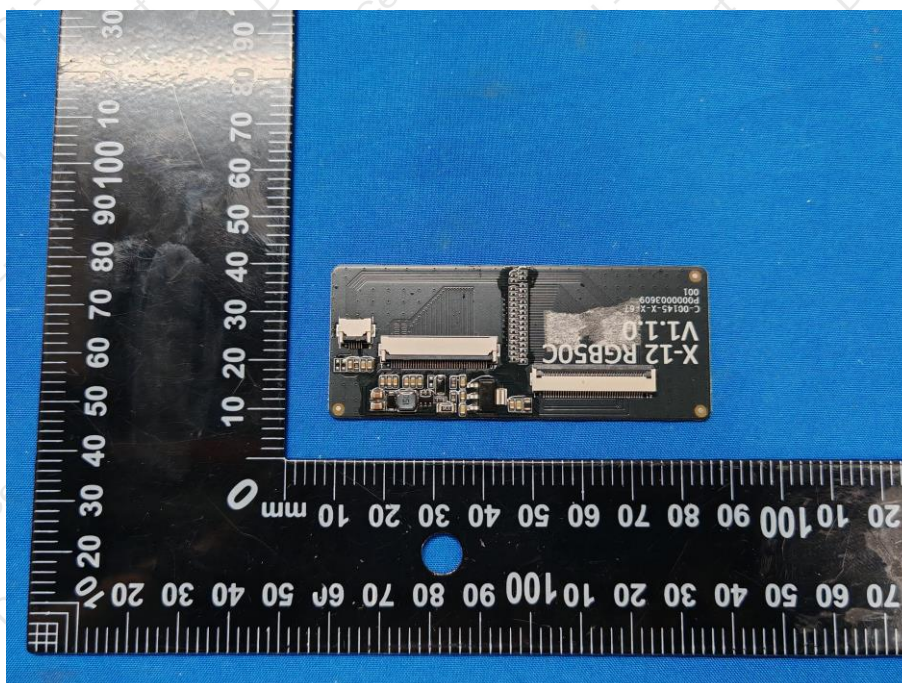


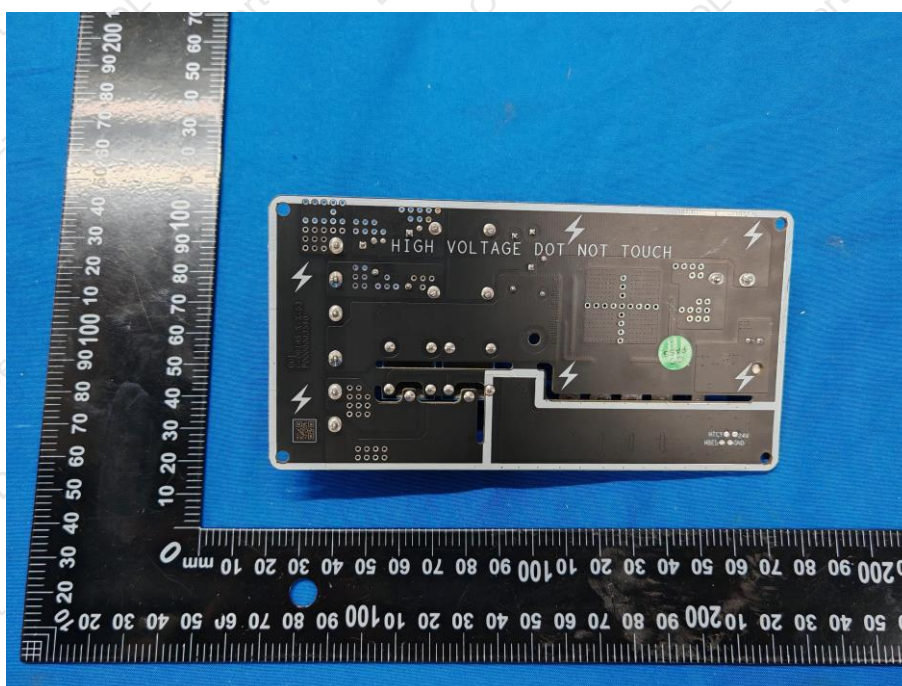
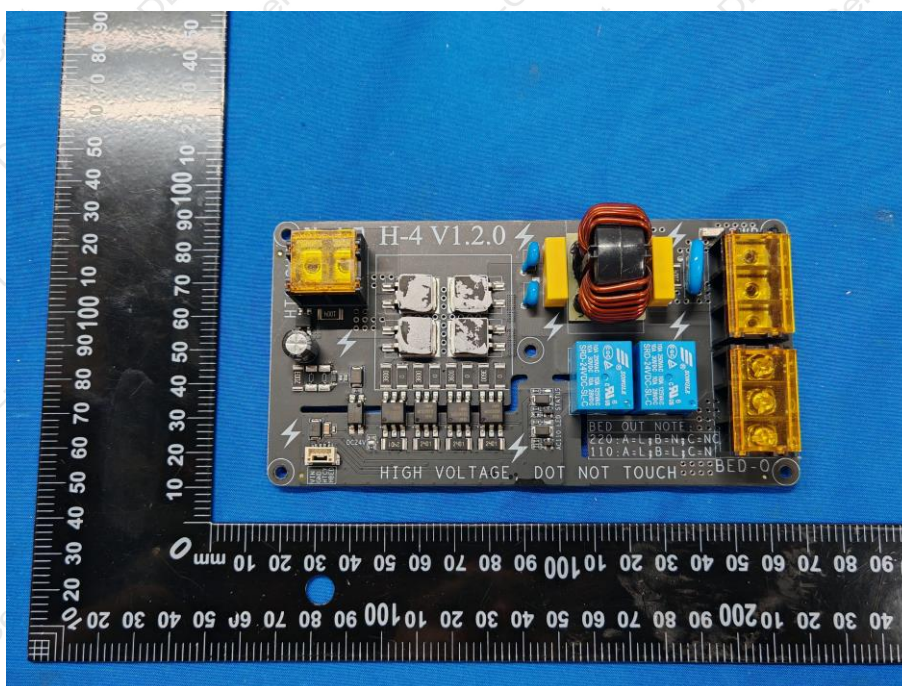


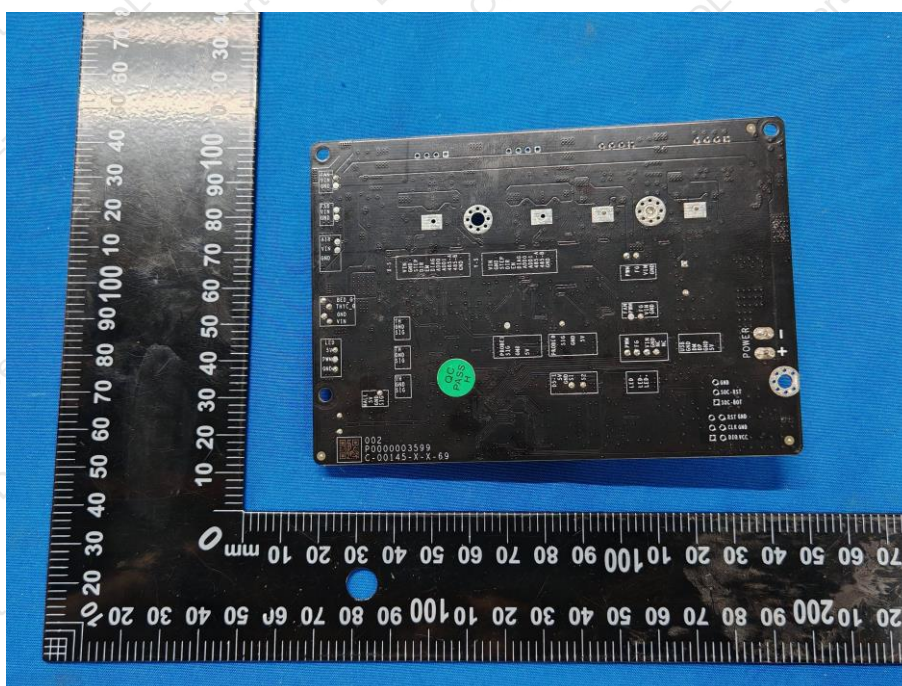
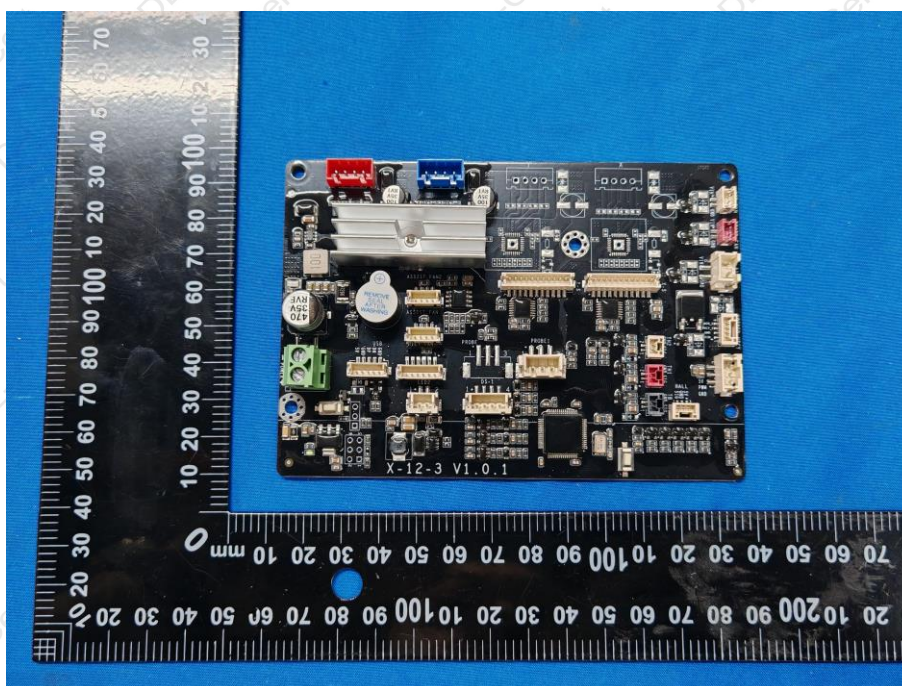


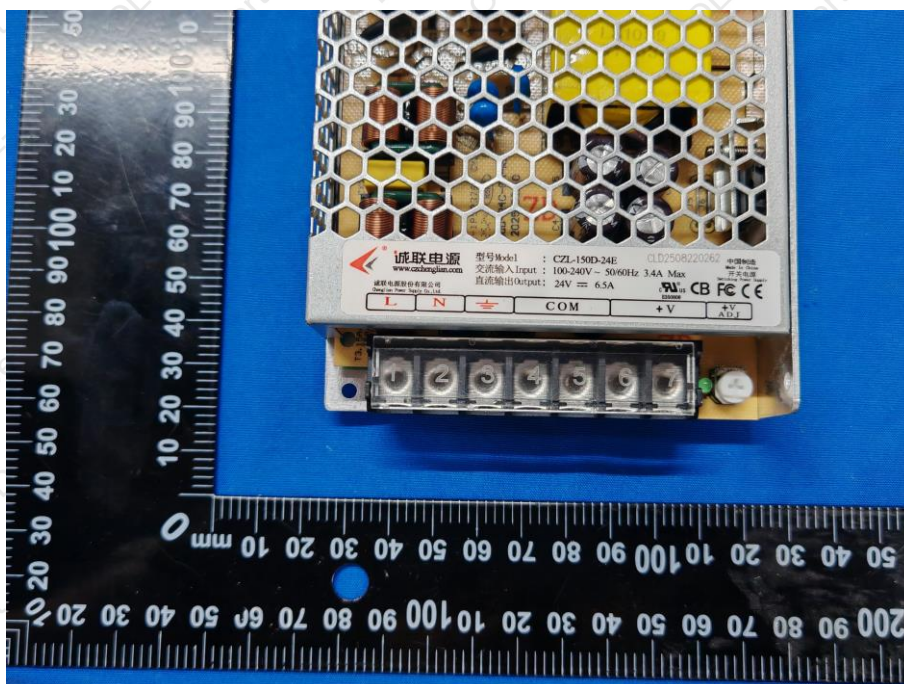
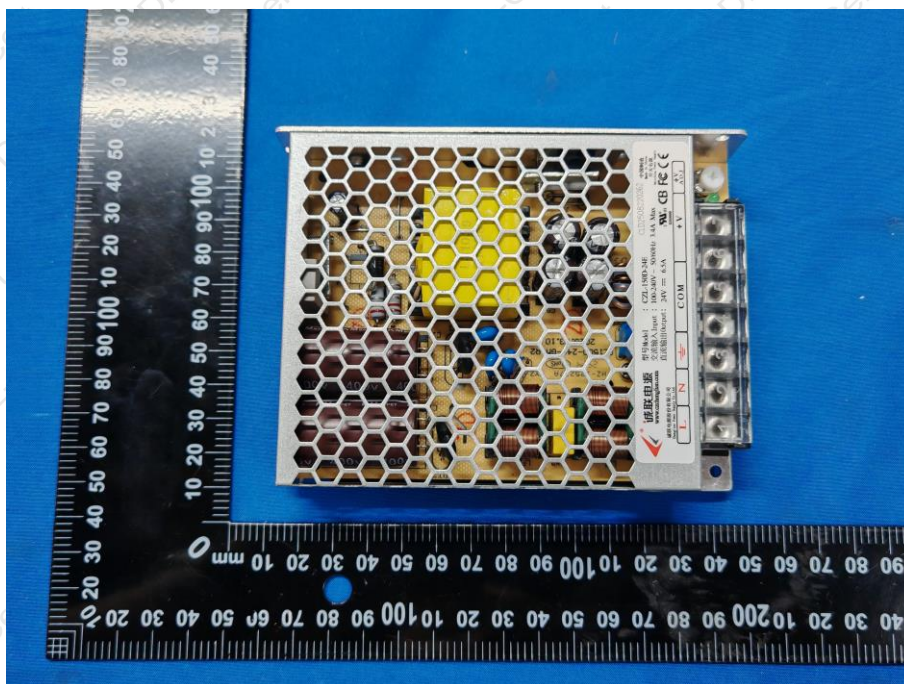


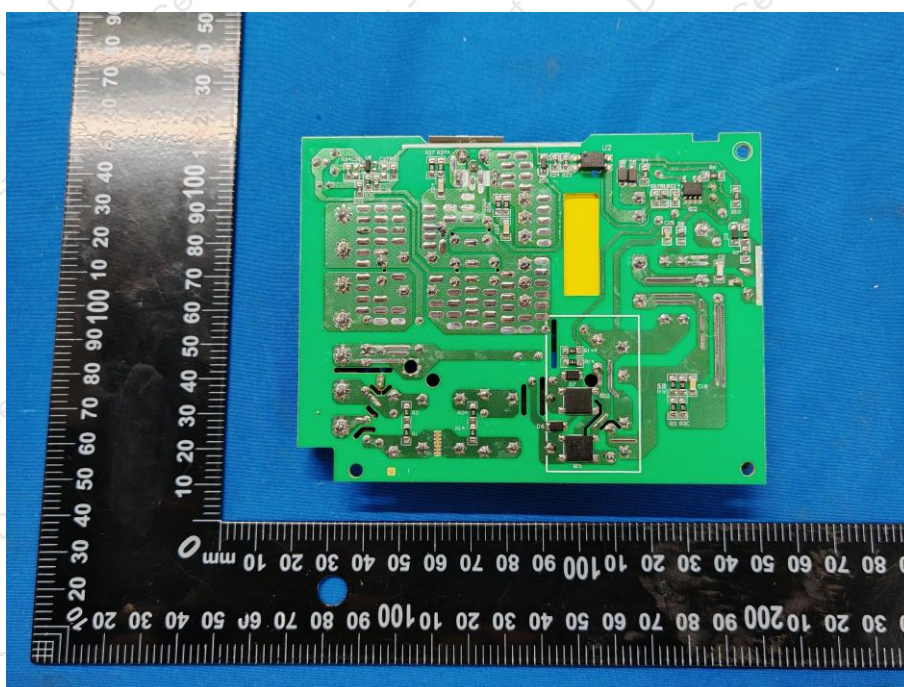
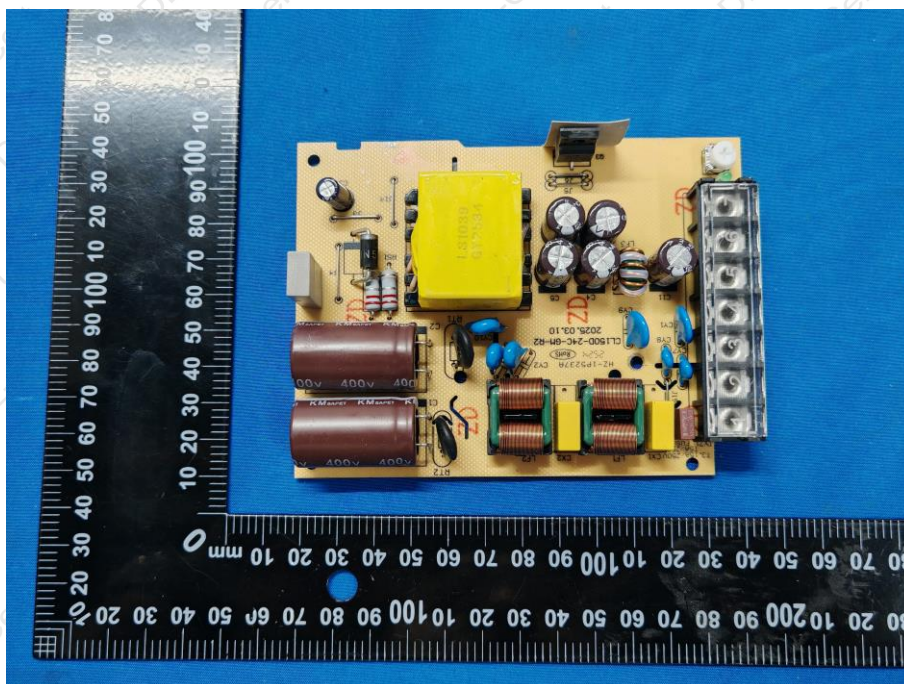


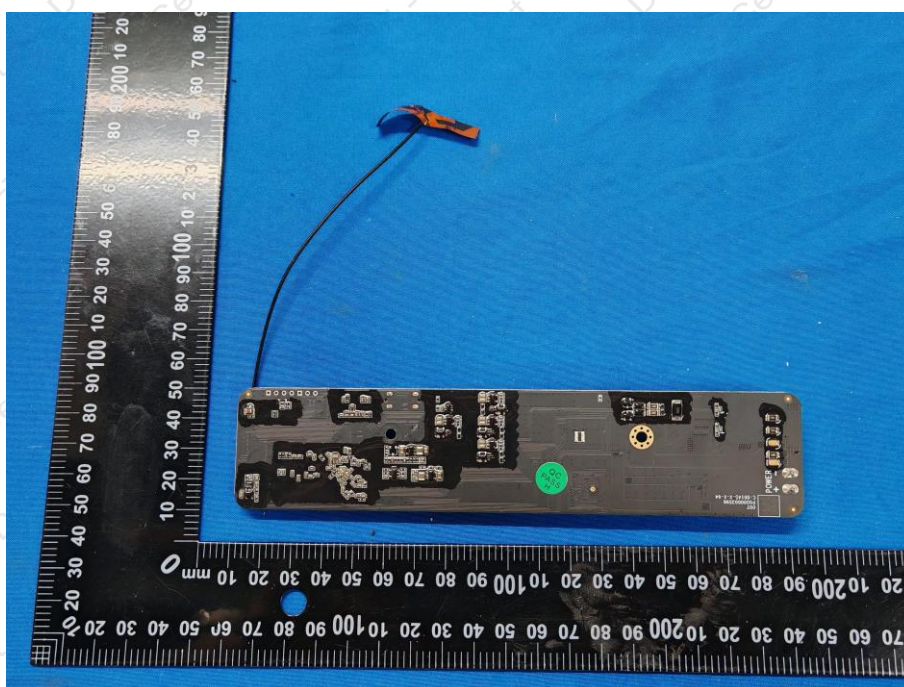
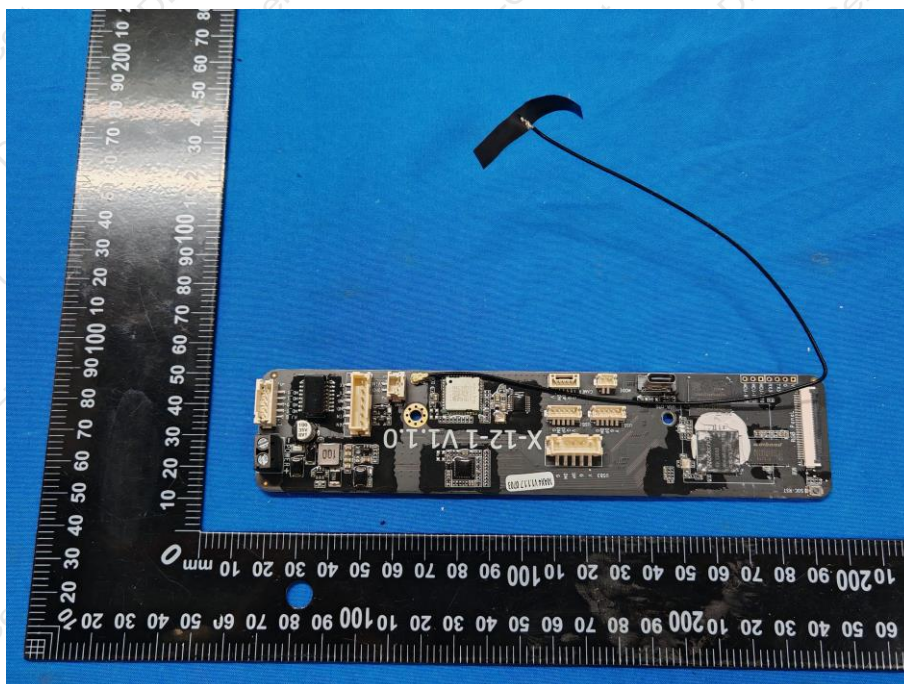


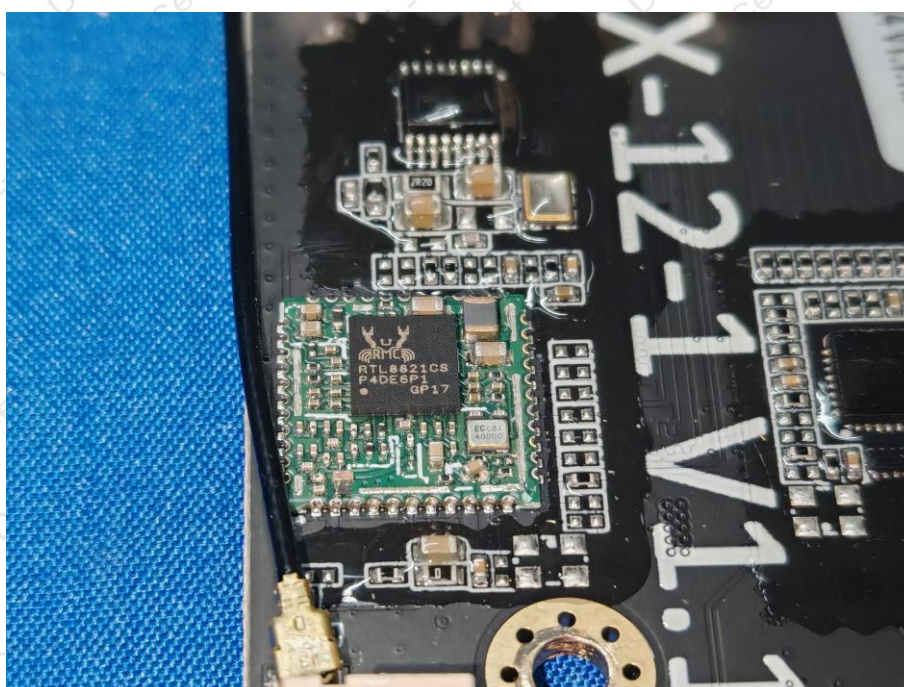
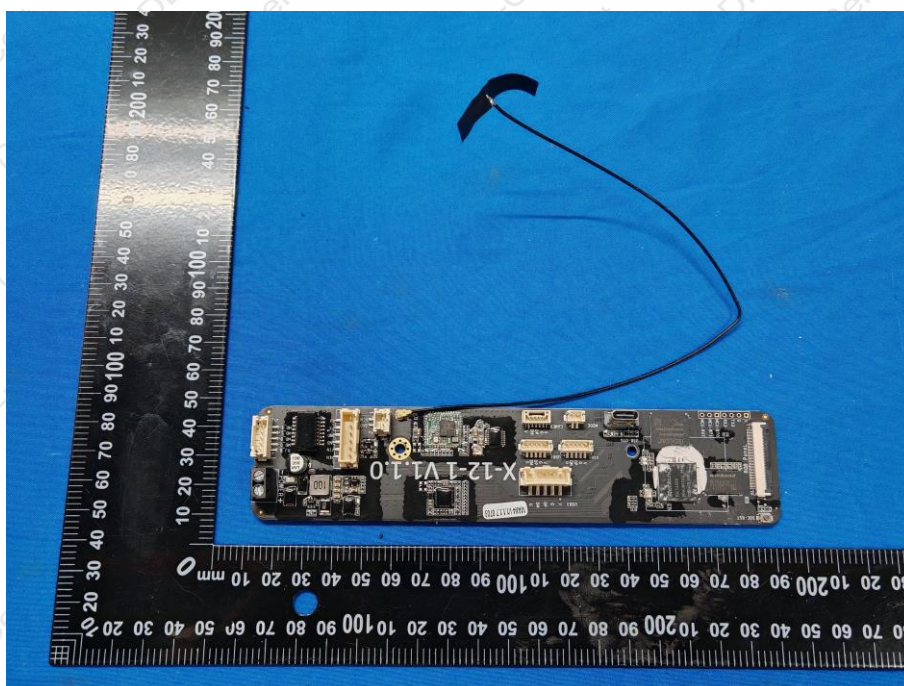


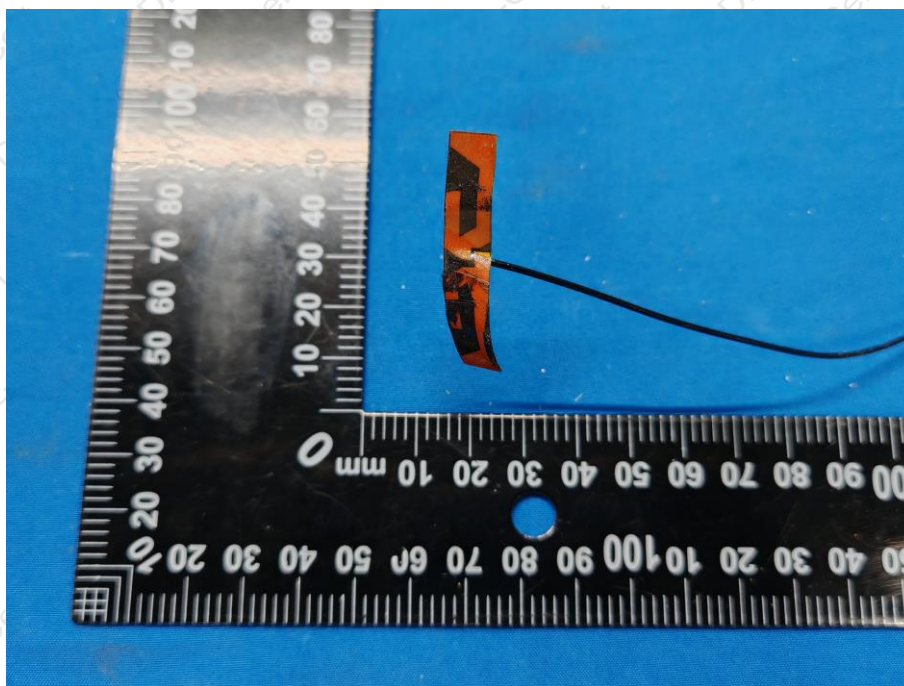












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